FACILITY RESPONSE PLAN

FOR THE DISCHARGE OF OIL OR HAZARDOUS SUBSTANCES AND OIL

FORT CAVAZOS, TEXAS

Prepared by:
DPW Environmental Division
4622 Engineer Dr.
Fort Cavazos, TX 76544

June 2023

(This document supersedes the Installation Response Plan dated June 2017)

RESPONSE PLAN COVER SHEET

U.S. Environmental Protection Agency (EPA) regulation, 40 CFR 112, Appendix F, requires that all owners or operators of facilities regulated under this part that pose a threat of substantial harm to the environment by discharging oil into or on navigable waters or adjoining shoreline are required to prepare facility-specific response plans. This plan follows the model facility-specific response plan guidelines found in 40 CFR 112, Appendix F. The cover sheet (below) must accompany the response plan to provide the EPA with basic information concerning the facility.

GENERAL INFORMATION

Owner/Operator of Facility: United States Army

Facility Name: III Armored Corps and Fort Cavazos

Facility Address (street address): 1001, 761st Tank Battalion, Fort Cavazos, TX 76544

Facility Phone Number: (254) 287-1110

Latitude (North): 31° 07" 51' Longitude (West): 97° 46" 04'

Dun & Bradstreet Number: N/A

Largest Aboveground Oil Storage Tank Capacity (Gallons): 636,488

Number of Aboveground Oil Storage Tanks: 600 [1]

North American Industrial Classification System (NAICS) Code: 928110[2]

Maximum Oil Storage Capacity (Gallons): 4,815,112

Worst Case Oil Discharge Amount (Gallons): 636,888

Facility Distance to Navigable Water. Mark the appropriate line:

0–1/4 mile __X__ 1/4–1/2mile ___ 1/2–1mile ____ >1mile ____

APPLICABILITY OF SUBSTANTIAL HARM CRITERIA

1. Does the facility transfer oil over-water[3] to or from vessels and does the facility have a total oil storage capacity greater than or equal to 42,000 gallons?

YES ______ NO __X__

2. Does the facility have a total oil storage capacity greater than or equal to one million gallons and, within any storage area, does the facility lack secondary containment[3] that is sufficiently large to contain the capacity of the largest aboveground oil storage tank plus sufficient freeboard to allow for precipitation within any aboveground oil storage area?

YES ______ NO __X__

3. Does the facility have a total oil storage capacity greater than or equal to one million gallons and is the facility located at a distance[3] (as calculated using the appropriate formula in Appendix C to 40 CFR 112 or a comparable formula) such that a discharge from the facility could cause injury to fish and wildlife and sensitive environments[4]?

YES __X__ NO ______

4. Does the facility have a total oil storage capacity greater than or equal to 1 million gallons and is the facility located at a distance[3] (as calculated using the appropriate formula in Appendix C to 40 CFR 112 or a comparable formula) such that a discharge from the facility would shut down a public drinking water intake?[3]

YES ______ NO __X__

5. Does the facility have a total oil storage capacity greater than or equal to 1 million gallons and has the facility experienced a reportable oil spill[3] in an amount greater than or equal to 10,000 gallons within the last 5 years?

YES ______ NO __X__

[1] Does not include a variable number of mobile and portable containers or oil-filled operational equipment. See Appendix C for additional information.

[2] These numbers may be obtained from public library resources.

[3] Explanations of the above-referenced terms are found in Appendix C to 40 CFR 112. If a comparable formula to the ones contained in Attachment C-III is used to establish the appropriate distance to fish and wildlife and sensitive environments or public drinking water intakes, documentation of the reliability and analytical soundness of the formula must be attached to this form.

[4] For further description of fish and wildlife and sensitive environments, see Appendices I, II, and III to NOAA's Guidance for Facility and Vessel Response Plans: Fish and Wildlife and Sensitive Environments (see Appendix E to 40 CFR 112, section 13, for availability) and the applicable ACP.

CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and that based on my inquiry of those individuals responsible for obtaining information, I believe that the submitted information is true, accurate, and complete.

COL Lakicia Stokes
Commanding
Fort Cavazos

23 JAN 24
Date

TABLE OF CONTENT

LIST OF ACRONYMS

The reference table below contains acronyms that are found throughout this plan.

Acronyms			
AAFES	Army and Air Force Exchange Service	**MIPR**	Military Interdepartmental Purchase Request
ACP	Area Contingency Plan	**MSC**	Major Subordinate Command
AF	Alert Facility	**NBC**	Nuclear, Biological, Chemical
AFSBn-Cavazos	Army Field Support Battalion – Fort Cavazos	**NCP**	National Contingency Plan
API	American Petroleum Institute	**NFC**	Nort Fort Cavazos
AR	Army Regulation	**NHWAP**	Non-Hazardous Waste Accumulation Point
ArcIMS	Arc Internet Map Server	**NPDES**	National Pollutant Discharge Elimination System
AST	Aboveground Storage Tank	**NRC**	National Response Center
BFSF	Bulk Fuel Storage Facility	**NRS**	National Response System
BLORA	Belton Lake Outdoor Recreation Area	**NSCC**	National Scheduling Coordinating Committee
BOA	Basic Ordering Agreement	**OPA 90**	Oil Pollution Act of 1990
CFR	Code of Federal Regulations	**OSC**	On-Scene Coordinator
CU	Classification Unit	**OSHA**	Occupational Safety and Health Administration
DFSP	Defense Fuel Support Point	**OSRO**	Oil Spill Removal Organization
DLA	Defense Logistics Agency	**OWS**	Oil/Water Separator
DLADS	Defense Logistics Agency Disposition Services	**PD**	DES Police Department
DOD	Department of Defense	**PM**	Provost Marshall (DES)
DPW	Directorate of Public Works	**POL**	Petroleum, Oils, and Lubricants
EDRC	Estimated Daily Recovery Capacity	**PREP**	National Preparedness for Response Exercise Program
EMS	Emergency Medical Service	**PVC**	Polyvinyl Chloride
EOC	Emergency Operations Center	**QI**	Qualified Individual
EPA	Environmental Protection Agency	**RCRA**	Resource Conservation and Recovery Act
ERAP	Emergency Response Action Plan	**RGAAF**	Robert Gray Army Air Field
F-24	Jet Fuel	**RGAF**	Robert Gray Alert Facility
FARP	Forward Arming and Refuel Point	**RRF**	Rapid Refuel Facility
FES	DES Fire and Emergency Services	**SCU**	Secondary Containment Unit
GDP	Gallons Per Day	**SDS**	Safety Data Sheet

HAZMAT	Hazardous Materials	**SERC**	State Emergency Response Commission
HAZWOPER	Hazardous Waste Operations and Emergency Response	**SPCCP**	Spill Prevention, Control, and Countermeasure Plan
HDPE	High-density Polyethylene	**SUPSALV**	U.S. Navy Supervisor of Salvage
IC	Incident Commander	**TCEQ**	Texas Commission on Environmental Quality
IOC	Installation Operations Center	**U.S.**	United States
IOSC	Installation On-Scene Spill Commander	**USCG**	U.S. Coast Guard
IRT	Installation Response Team	**UST**	Underground Storage Tank
LEPC	Local Emergency Planning Committee	**WFC**	West Fort Cavazos
MC	Main Cantonment	**YDAHP**	Yoakum-Defrenn Army Heliport
MEDDAC	Medical Department Activity		

1. EMERGENCY RESPONSE ACTION PLAN (ERAP)

The ERAP has been compiled for easy access by response personnel during an emergency. Collectively, the procedures presented below describe the initial response actions following a worst-case discharge. These actions include stopping the source of the spill, notifying the appropriate people, and preventing or minimizing the spread of fuel.

1.1 QUALIFIED INDIVIDUAL (QI) INFORMATION

The basic duties of the QI include the following: Daily operations management; spill identification; immediate response to a spill (within two (2) hours of spill event); local, state, and federal agency notification; coordination of response operations; and liaison responding agencies. An emergency event requires immediate notification of the QI or alternate(s). The contact information for the QI and alternates are listed below:

QI Name: Timi Dutchuk
Position: Chief, Environmental Division
Work address: 4622 Engineer Dr, Fort Cavazos, TX 76544-5021
Emergency phone number: (254) 760-5636

Alternate QI Name: Sean Goodwin
Position: Chief, Environmental Management Branch
Work address: 4622 Engineer Dr, Fort Cavazos, TX 76544-5021
Emergency phone number: (254) 535-1007

1.2 REPORTING AND NOTIFICATION REQUIREMENTS

1.2.1 NATIONAL RESPONSE CENTER (NRC) CRITERIA

The NRC notification requirements are outlined in Code of Federal Regulations (CFR), Title 40, Part 110 Discharge of Oil: §110.3 Reportable Spill. Discharge of oil in such quantities that the administrator has determined may be harmful to the public health or welfare or the environment of the United States include discharges of oil that: (1) Violate applicable water quality standards; or (2) Cause a film or sheen upon or discoloration of the surface of the water or adjoining shorelines or cause a sludge or emulsion to be deposited beneath the surface of the water or upon adjoining shorelines.

1.2.2 TEXAS COMMISSION ON ENVIRONMENTAL QUALITY (TCEQ) CRITERIA

The TCEQ reporting requirements are outlined in Texas Administrative Code (TAC), Title 30, Chapter 334 Underground and Above Ground Storage Tanks §334.75 Reporting and Cleanup of Surface Spills and Overfills.

a. Owners and operators of aboveground storage tanks (AST) and underground storage tank (UST) systems must contain and immediately clean up a spill or overfill, report the spill or overfill to the agency within 24 hours, and begin corrective action of any spill or overfill of petroleum substance from an UST or any spill or overfill of petroleum product from an AST that results in a release to the environment that exceeds 25 gallons, or that causes a sheen on nearby surface water.

b. Owners and operators must contain and immediately clean up a spill or overfill of any petroleum substance from an UST or any petroleum product from an AST that is less than 25 gallons. If cleanup cannot be accomplished within 24 hours, owners and operators must immediately notify the agency.

1.2.3 REPORTING RESPONSIBILITY

The DPW Environmental Division (ENV) will be responsible for making notification to regulatory agencies and the ensuring IC is aware of all notifications.

1.2.4 EMERGENCY NOTIFICATION PHONE LIST

Table 1-1, Emergency Notification Phone List, provides the names and phone numbers of the organizations and personnel for immediate notification by the DPW Environmental Division.

Table 1-1 Emergency Notification Phone List	
Federal and State Organizations	**Phone Number**
National Response Center **(NRC)**	(800) 424-8802
Federal On-Scene Coordinator – EPA Region 6 - 24-hour spill reporting Region 6 Emergency Response Center	(800) 424-8802 (866) 372-7745
Texas Commission on Environmental Quality **(TCEQ)** – Primary and Alternate Spill Emergency Spill Response	(254) 744-6699 (254) 744-4741
State On-Scene Coordinator – TCEQ - 24-hour spill reporting Region 9 Office, Waco, TX (M-F 8-5)	(800) 832-8224 (254) 751-0335
State Emergency Response Commission (SERC) SERC 24-hour number SERC Alternate 24-hour number	(800) 832-8224 (512) 463-3199
Texas State police	(512) 465-6261
National Weather Service Forecast Office, Fort Worth, TX	(817) 429-2631
Local Community Organizations	**Phone Number**
Local Emergency Planning Committee **(LEPC)**: Bell County LEPC Evenings: Sheriff's office Coryell County LEPC Evenings: Sheriff's office	 (254) 933-5587 (254) 933-5412 (254) 223-4123 (254) 865-7201
Local Police Station	911
Bell County Water Control and Improvement District No. 1 Belton Lake Water Treatment Plant (normal business hours) 24-hour phone	 (254) 526-6343 (254) 939-2481
Fort Cavazos Emergency Organizations	**Phone Number**
Fire and Emergency Services (FES) Fire Dispatch	911
Installation Operations Center (IOC)	(254) 287-2520

Military Police	(254) 288-1062
Public Affairs Office	(254) 286-5139
Fort Cavazos Directorate of Public Works (DPW) Organizations	**Phone Number**
Director of Public Works (normal duty hours)	(254) 287-5707
Work Reception (normal and after duty hours)	(254) 287-2113
Environmental Division: Fort Cavazos Environmental Division Front Desk Environmental Response Coordinator (normal duty hours) Environmental Response Coordinator (alternate/after duty hours)	(254) 287-6499 (254) 288-5284 (254) 535-3228
Fort Cavazos Health and Welfare Organizations	**Phone Number**
Darnall Army Community Hospital	(254) 288-8000
Ambulance	911
Fort Cavazos Industrial Hygiene	(254) 288-1664
Defense Logistics Agency (DLA)	**Phone Number**
DLA Energy and Staff Duty Officer .... DLA.spillreports@dla.mil DLA Energy Staff Duty Officer (24-hour)	Email Provided (517) 767-8420
Army Petroleum Center .... usarmy.belvoir.asc.list.apc-operations@army.mil	(517) 767-0661
DLA Energy America East ....DLA-AM.spillreports@dla.mil	(713) 718-3883
Maytag Operations Office (Fort Cavazos)	(254)-285-2170
Oil Spill Removal Organizations (OSRO)	
USA Environmental, LP (24-hour)	713) 425-6900
TAS Environmental (24-hour)	(888) 654-0111
US Ecology (24 hour)	(800) 899-4672

1.3 SPILL RESPONSE NOTIFICATION FORM

The FRP Form 1-1, Spill Response Notification Form, is a checklist of information that will be provided to the TCEQ, NRC, or any regulatory agency as required.

1.3.1 INITIAL NOTIFICATION REQUIREMENTS

a. TCEQ Initial Notification Requirements – Initial notification will be made when all readily available information has been collected and documented on the form by the Incident Commander (IC) or Installation On-Scene Coordinator or other designated emergency personnel, no later than 24 hours after discovery. Any missing information will be reported in a timely manner as it is collected.

b. NRC Initial Notification Requirements - 40 CFR 110.6 requires that any person in charge notify the NRC immediately upon gaining knowledge of the discharge of oil from the facility. The QI (identified in Section 1.1 above) will conduct notifications. The NRC will be notified immediately after the QI has gained knowledge of the discharge.

1.3.2 DLA ENERGY NOTIFICATION

If the spill is from one of the DLA Energy bulk tanks, additional reporting requirements apply. As per paragraph 3.5 of the DLA Policy Letter P-40, if a spill from DLA facilities is of 25 gallons or greater on land or in an amount that causes a sheen on water, the facility must report the spill to DLA-WE, the Service Control Point, and the DLA Region as quickly as possible, but not later than 24 hours from discovery. Reporting method to DLA-WE and the Staff Duty Officer is through email or by phone. DLA contact information is listed in Table 1-1 Emergency Notification Phone List (Section 1.2.3 of this document). The DLA-E's O&M contractor, Maytag Aviation Corporation, is responsible for making these notifications to DLA.

1.3.3 FRP FORM 1-1, SPILL RESPONSE NOTIFICATION FORM.

All information should be provided on the form at the time of notification; however, spill notification should not be delayed for information collection.

FRP Form 1-1 Spill Response Notification Form
Instructions: This Information should be collected as soon as possible during the initial site assessment phase by the Incident Commander, Installation On-Scene Coordinator, or another designated emergency response official, without endangering personnel. Once complete it should be provided to the DPW Environmental Division (Bldg. 4622) for regulatory reporting. **All information must be included on the form at the time of notification; however, spill notification shall not be delayed while collecting the information on the form.**
Section I Incident Description
Name and telephone number of reporter or reporting agency: Date/time of release and/or discovery: Location of spill: Type of incident: ☐ leak ☐ fire ☐ spill ☐ explosion ☐ transportation accident Cause, source and duration of spill (descriptive narrative): Container type and total capacity (truck, railcar, drum, tank, etc.):
Section II Injury/Fatality and Potential Danger
Number and type of injuries: Number of fatalities: Potential dangers (fire, explosion, toxic vapor, etc.):
Section III Environmental Conditions

Nearby Populations: Weather conditions: Wind direction: Terrain conditions (slope, surface type): Nearby sensitive environments (water bodies, vegetation, etc.):
Section IV Spill Classification and Identification.
Spill classification: ☐ Minor (oil > 1,000 gal; HAZMAT – minimal threat to public health/welfare) ☐ Medium (oil between 1,000 and 10,000 gal; HAZMAT – not classified as minor or major) ☐ Major (oil < 10,000 gal; HAZMAT – substantial threat to public health/welfare) Specific description or identification of the oil, petroleum product, hazardous substances or other substances discharged or spilled: Material identifying marks (manifests, placards, labels, ID and/or NFPA numbers, etc.): Characteristics of Spill (odor, color, gas, liquid, solid, etc. if detectable):
Section V. Environment Impacts
Name and description of waters affected or threatened by the discharge of spill: Description of actual or potential water pollution or harmful impacts to the environment (list sensitive areas and/or natural resources at risk potentially threatened by spill): Description of any known or anticipated human health risks:
Section VI. Response Actions
Does spill meet reporting criteria? Yes ☐ No ☐ Did spill enter storm/sewer system? Yes ☐ No ☐ Was spill contained to immediate area? Yes ☐ No ☐ Was Fort Cavazos Fire Department notified? Yes ☐ No ☐ Was dig permit requested? Yes ☐ No ☐ Was product extracted? Yes ☐ No ☐ Amount: Was soil excavated? Yes ☐ No ☐ # of cubic yards: A description of any additional actions that have been taken, are ongoing, or will be taken to contain and respond to the discharge or spill:

1.4 RESPONSE EQUIPMENT AND LOCATION

Fort Cavazos maintains a large quantity of spill response materials and equipment at strategic locations across the installation to be easily accessed by spill responders.

1.4.1 SPILL EQUIPMENT LIST BY LOCATION

Most of the large spill equipment is stored at the Pollution Prevention (P2) Services facility. Table 1-2, Spill Equipment List by Location, identifies the organizations and locations where equipment is stored; the type of equipment at each location; and the primary mission of that organization as it relates to spill response. A full inventory of spill response equipment is maintained and updated by the DPW ENV.

1.4.2 TYPE OF EQUIPMENT

Emergency response equipment owned or operated by the Installation, as outlined in Table 1-2, are available on-site in the event of a discharge or release. Response equipment and materials stored on Fort Cavazos include:

a. Standard spill response kit materials: absorbent pads (general purpose, oil and acid), small absorbent booms, and sweepable absorbent

b. Small response equipment: non-sparking shovels, brooms, pop-up containment, curb inlet diversion dams, storm drain covers, plastic bags, and over-pack drums

c. Large spill response equipment: Containment creek boom with trailer, hazmat trailers, skimmers, culvert plugs, and large absorbent booms

d. Spill response vehicles and Storage Tanks: Spill response trucks, VAC and skimmer trucks, and aboveground storage tank

Table 1-2 Spill Equipment List by Location

Organization	Location	Equipment	Mission
Military Maintenance Facilities	Across the Installation	Standard spill response kits as required by the Fort Cavazos SPCC Plan	For use with MFT and MFS operations
DPW ENV	Bldg. 4622)	Standard spill kit materials Small response equipment (loaded on work vehicles)	Incidental spills from equipment/vehicle leaks, training exercises in range/TAs, and fuel transfer.
DPW O&M General Support Shop	Bldg. 40001	Standard spill kit materials	Management of used POL products Fuel transfer
DPW O&M Roads and Grounds	Bldg. 40001	*Earth Moving and Soil Excavation Equipment	Clean path of recovery and erect earthen containment dams
LRC Maintenance Facility	Bldg. 88030	Standard spill kit materials	Management of used POL products Fuel transfer
BLORA	Bldg. NAF126	Standard spill kit materials	

FES Fire Station 2	Bldg. 90145	HAZMAT trailers	Emergency spill Response
DPW ENV P2 Services Facility	Bldg. 1948	Response trailer: Standard spill kit materials, Small response equipment; Large absorbent booms Culvert Plugs Creek boom with trailer *VAC/Skimmer Trucks *ASTs for storage of product (approximately 10,000)	Environmental response Spill containment Removal and storage of free product

**Note: Availability of equipment such as work vehicles, VAC/Skimmer Trucks, ASTs, and earth moving equipment that are not dedicated solely to emergency spill response is dependent on mission requirements and operational status at the time of the spill and may be limited.*

1.5 RESPONSE EQUIPMENT TESTING AND DEPLOYMENT

This section pertains to equipment purchased and maintained for the sole purpose of spill response. An inventory of that equipment is maintained by DPW ENV.

1.5.1 EQUIPMENT MAINTENANCE

Spill response equipment is inspected and serviced at least annually or in accordance with the manufacturer's recommendations and best commercial practices to ensure that equipment is available for deployment in the event of emergency release or discharge, routine incidental spills, and training exercises. Any equipment in need of maintenance will be recorded as temporarily unavailable until maintenance can be completed. Any equipment that exceeds shelf-life or is determined to be no longer serviceable for its intended use, will be either turned in for disposal or marked as "For Training Use Only" and removed from available inventory. The equipment will be replaced as quickly as funds are available. Equipment inspection and service records will be maintained by the owning organization and made available when requested.

1.5.2 EQUIPMENT TESTING AND DEPLOYMENT

Fort Cavazos spill response equipment is periodically deployed during scheduled and unscheduled training drills/ functional exercises, and during routine incidental spills. All emergency or routine incidental spills are annotated in a spill response log along with a detailed record for each spill. Training drills/exercises are recorded in the spill response log (identified as drill/functional exercise) and records will be maintained in a corresponding electronic file. Spill response log and corresponding records are maintained by the DPW Environmental Division.

Note: The Fort Cavazos Fire and Emergency Services (FES) maintains their own equipment testing and deployment records.

1.6 SPILL RESPONSE PERSONNEL

The Fort Cavazos Fire and Emergency Services (FES) is the primary response agency for both emergency and routine incidental spills. The DPW Environmental Response Team will provide support when requested. If a release or discharge exceeds Fort Cavazos's response capabilities, a regional emergency response company will be requested as an Oil Spill Removal Organization (OSRO) for additional support. When collection of a hazardous product is required for proper disposal, FES personnel will request a representative from the DPW Environmental Response Team to respond to the scene to coordinate the transfer/disposal with the responsible party.

1.6.1 FORT CAVAZOS FIRE AND EMERGENCY SERVICES (FES)

The FES serve as the Installation Response Team (IRT) for Fort Cavazos and provides emergency support for fires, aircraft emergencies, release, or discharge of hazardous materials (HAZMAT), and other emergency incidents that occur on the installation. The FES maintains 32 firefighters on duty at all times, and approximately 117 firefighters available around-the-clock. The FES Dispatch can be contacted 24 hours a day by dialing 911 on any base telephone or (254) 287-3908.

a. All FES firefighters GS-07 level and above are trained at a minimum to the HAZMAT Technician Level I (Title 29 CFR 1910.120(q)(6)(iii)) and are Department of Defense (DoD)-certified first responders. The FES will be the first on scene to secure the site and identify and mitigate threats to human health or safety.

b. For routine incidental spills where no immediate threat is present, IRT personnel will respond and initiate the Incident Command System in accordance with the National Incident Management System (NIMS), attempt to control the release or discharge source, contain, and provide information to FES Dispatch. FES Dispatch will relay the information to the DPW ENV and request support from the DPW Environmental Response Team if needed.

c. If an immediate threat is present the FES will initiate the Incident Command System IAW NIMS and provide the EOC with required information. FES will remain in command and continue to provide support for as long as they are needed on the site. The IC will coordinate with the DES Police Department to provide traffic control and site security and/or initiate evacuation procedures as required.

d. The FES also has Mutual Aid Agreements in place with local jurisdictions.

1.6.2 DPW ENVIRONMENTAL RESPONSE TEAM

The DPW Environmental Response Team is comprised of DPW ENV personnel from different program, who are trained and qualified to provide services in their specialty program areas. Personnel whose duties qualify, are also trained to the First Responder Operation Level (Title 29 CFR 1910.120(q)(6)(ii). DPW Environmental Division hours are Monday – Friday from 7:00am to 4:30. DPW Environmental Personnel will respond after duty hours and on weekends as requested by DES Dispatch, but do not have the capabilities or manpower to work 24hr operations.

a. For routine incidental spills where no immediate threat is present, the Environmental Response Team will assist the FES with containment and provide support to the spill organization during cleanup activities.

b. If an immediate threat is present and an incident command is required, Once notified of a release or discharge by FES Dispatch, the IOSC/alternate IOSC will notify the QI/alternate QI, inform them of the situation, and initiate the Environmental Response Team response. The Environmental Response Team will report to the equipment P2 Services to prepare equipment for deployment. The IOSC will proceed to the incident command location to assist the IC, request additional installation support if required, and relay information to the Environmental Response Team with deployment equipment requirements and location(s).

c. Initial DPW ENV response time is typically between 20 and 45 minutes depending on the location of the spill. Once a representative is on-site, the situation will be assessed, and the Environmental Response Team with appropriate resources can be deployed within 1 to 1 ½ hours.

d. Once containment has been achieved, IOSC will coordinate with other Environmental Response Team personnel for free product recovery and storage or environmental impacts assessment.

e. Table 1-3, DPW Environmental Response Team, below provides information for each program area with a brief description of services provided and response training required in that program area.

Table 1-3 DPW Environmental Response Team
24-Hour DPW Emergency Response Number: (254) 535-3228

Program	Primary Mission Role and Training	Training
Clean Water Program	Provide IOSC to IC to coordinate response efforts, collect reporting information, and keep QI informed. Make regulatory notifications. Lead down-stream equipment deployment operations for containment if necessary	24 Hour HAZWOPER 8 Hour HAZWOPER Annual Refresher ICS Incident Command Training
Pollution Prevention Services	Conduct free product recovery and storage operations within equipment, manpower, and storage capabilities.	24 Hour HAZWOPER 8 Hour HAZWOPER Annual Refresher
Hazardous Waste Program	Coordinate and receive contaminated wastes for disposal through the Classification Unit.	RCRA Annual Training
Natural Resources	Assess and record information concerning environmental impacts on vegetation, fish, and wildlife at and around spill site.	No specific spill response training required

Note: A list of the DPW Environmental response personnel and contact information is maintained by the IOSC and available to Fort Cavazos emergency response personnel.

1.6.3 CONTRACTOR RESPONSE

The Defense Fuel Support Point (DFSP) at Fort Cavazos is a government-owned, contractor-operated facility that stores and supplies bulk fuel to military units at Fort Cavazos. The DFSP contractor, Maytag Aircraft Corp., operates under a Defense Logistics Agency (DLA) contract. The contractor operates DLA Energy's capitalized product facilities on the installation. However, the U.S. Army Garrison Fort Cavazos maintains ownership of all real property at these fuel facilities.

a. DFSP Contractor Support: For incidental spills at DLA contract operated facilities, Maytag is responsible for stopping and cleaning up the spill with Fort Cavazos response personnel support. Maytag maintains spill supplies at each site for immediate containment and cleanup of routine leaks and small spills. For larger spills the FES, would be notified and Fort Cavazos response procedures as outlined above would be initiated. If the release or discharge exceeds Fort Cavazos's response capabilities, as determined by the QI, an OSRO will be requested by Fort Cavazos for additional support.

b. OSRO Response: The emergency response company contracted to act as OSRO is responsible for the immediate spill containment, cleanup and recovery of free product. Once the initial cleanup and recovery of free product is completed, the cleanup actions will switch to a remediation phase, which could include long term monitoring and further cleanup actions if necessary. Once requested, the regional OSRO support can respond within Tier 1 times (according to 33 CFR Part 154, Subpart F, Tier times are: Tier 1, 12 hours; Tier 2, 36 hours; Tier 3, 60 hours for Fort Cavazos geographic region). If additional resources are needed, U.S. Coast Guard (USCG) basic ordering agreement (BOA) emergency response contractors are located in several cities within Texas. Regional emergency response companies are listed below in Table 1-4.

Table 1-4 Emergency Response Contractors

Name and Phone Number	Response Time	Response Capabilities
TAS Environmental, LP (888) 654-0111	2.0 hours Austin 4.5 hours from D/FW or San Antonio	**Worst-case Discharge Responder** TAS offers a full suite of environmental compliance services including consulting, training, and emergency response. With multiple offices throughout Texas, TAS is positioned to offer full cleanup solutions.
USA Environment, LP (281) 996-4373	3.0 hours from New Braunfels 6.5 hours from Houston	**Worst-case Discharge Responder** USA Environment is a full service environmental compliance and emergency response company including spill response, remediation, waste management, storage tank cleaning, training, and many other services. USA operates several offices throughout Texas and is very capable of providing emergency response cleanup services.
US Ecology (800) 899-4627	4.5 hours from D/FW or San Antonio	**Worst-case Discharge Responder** US Ecology is a full-service environmental compliance and emergency response company with multiple locations in Texas. Services include spill response, remediation, and waste management.

1.7 EVACUATION PLAN

All government and contract organizations on the installation are required to develop and implement an Emergency Action Plan (EAP) for execution in case an emergency. The EAP provides guidance and instruction for personnel to execute in the event of a broad variety of emergency scenarios. One of the requirements of the EAP is to provide site evacuation procedures for each building in the organization's area of operations.

1.7.1 SITE EVACUATION PROCEDURE

The site evacuation procedures are to be followed in the event of an emergency evacuation. The EAP provides information on the organization alarms or notification systems, location of personnel rosters, and facility maps identifying exits, paths of egress, and assembly areas. Signs are posted within the building to remind occupants of evacuation procedures and re-assembly areas. If the alarm or emergency notification system is activated in a building or on a site, organization personnel will evacuate the building according to the plan and proceed to a pre-established assembly areas for accountability by supervisors or building managers. Supervisors and/or building managers or the senior organization person on the site are responsible for ensuring that personnel are present, or their location is verified, and providing the accountability information to emergency personnel when they arrive on site.

1.7.2 NOTIFICATIONS SYSTEMS

The buildings on the installation are equipped with fire/emergency alarm system which provide notification to building personnel to evacuate the building. If the building does not have an alarm system, an alternate method to alert building personnel of an emergency, must be identified in the site evacuation plan. In addition to individual building alarms, Fort Cavazos has several forms of emergency broadcasting capabilities, includes notifications by: Installation siren or "Giant Voice" Alert! Information Hotline (254) 287-6700, Fort Cavazos Website; land line telephone; email notifications for Fort Cavazos email addresses, mobile phone application and local news and radio stations.

1.7.3 ARRIVAL AND EVACUATION ROUTES

The DES Police Department in coordination with the IC will evaluate all of the variables before determining the safest arrival and evacuation route. Some of the variables for consideration are Spill site information provided by the IC; number of evacuees; potential for hazard to spread; weather conditions and wind direction/velocity; current road closures; available transportation; resources from adjacent cities; and adequate maneuverability for emergency response vehicles.

1.7.4 COMMAND CENTER AND TEMPORARY SHELTER

The need for a centralized check-in and temporary shelter locations will be determined and activated by the IC, if an immediate threat of fire, explosion, or exposure is present and/or expected to persist for a prolonged period of time. Temporary staging areas and shelters will be established to accommodate evacuees. The installation has a shelter plan that the IC directs.

1.7.5 MEDICAL ATTENTION

After initial decontamination in the field, injured personnel would be transported to the closest medical facility as determined by EMS.

1.8 IMMEDIATE ACTION

If a release or discharge occurs or is discovered, the spilling or discovering organization is required to notify the FES and the site supervisor or commanding officer. The IRT provides initial response to calls for POL discharges or spills reported to the Fire Dispatch office or Range Operations. Once on site, IRT personnel will assess the site for any threats of fire, explosion, or exposure. An Immediate Action Spill Response Poster is located at Appendix G of the Fort Cavazos SPCCP and is available for reproduction and display.

1.8.1 RESPONSE OF SPILLING OR DISCOVERING ORGANIZATION

The spill site supervisor/commanding officer will perform the following actions only if it can be done safely without further endangering personnel:

a. Identify the source and amount of the spill/release (if it can be performed safely).

b. Evaluate the spill (Section 5.5 of this plan) and ensure that the FES is notified for spills over 5 gallons (or entering the environment).

c. Activate the site's emergency alarm system if evacuation is necessary.

d. Initiate evacuation.

e. Ensure the following immediate actions are taken if they can be performed safely and do not endanger personnel: Stop or slow the flow of product, shut off any ignition sources, and Initiate containment.

f. For spills that are under reporting limits, initiate clean-up.

1.8.2 NO INDICATION OF IMMEDIATE THREAT

Once on site, if the IRT determines there is no immediate threat to human health and safety present they will initiate the following spill response activities:

a. IRT will attempt to control or stop the source, initiate containment, and provide information to FES Dispatch.

b. Dispatch will notify DPW ENV (IOSC) to provide information and request support.

c. The IOSC/Alternate IOSC will provide information to the QI/Alternate QI and contact the Environmental Response Team personnel needed to respond and assist the IRT containment efforts.

d. FES will maintain control of the site until DPW Environmental Personnel have been fully briefed, a positive battle hand-off has occurred, and can assume responsibility for monitoring site cleanup activities.

e. The Environmental Response Team will remain on site when the IRT mission is complete to provide support to spilling organization and monitor cleanup activities to ensure the site is returned to pre-spill conditions.

f. IRT will stay on site until DPW Environmental Response Team personnel have been briefed on situation.

g. If spill meets required reporting criteria, the IOSC or alternate will complete FRP 1-1, Spill Response Notification Form and make necessary notifications within the timeline requirements for each agency.

1.8.3 IMMEDIATE THREAT TO HUMAN HEALTH OR SAFETY

If the IRT determines there is an immediate threat present, the threat will be mitigated before further spill response activities are initiated. The IRT will report the threat to FES dispatch center and request support from the Directorate of Emergency Services (DES) and the following actions will be taken:

a. DES Police Department will conduct emergency evacuation, site security and traffic control if required.

b. FES will establish command of the site, designate an incident command site, notify the Installation Operation Center (IOC), determine an arrival route for emergency and response vehicles, and relay information to FES Dispatch for IOSC notification. The IRT will work to clear all threats, attempt to stop, or control the discharge source, and initiate containment procedures.

c. The IOSC will notify and brief the QI, alert the Environmental Response Team to prepare for response, and proceed to the incident command site to coordinate response efforts with the IC, provide Environmental Response Team members with information for response determinations, and collect information for FRP form 1-1, Spill Response Notification Form. The IOSC will maintain contact with the Environmental Response Team and the QI to record and relay any environmental actions and site information as soon as possible.

d. QI/alternate QI will notify and provide required information to regulatory agencies, request support from higher authorities if required, and request OSRO support if information provided indicates that spill exceeds Fort Cavazos personnel and equipment capabilities. The QI is responsible for escorting regulatory agencies that arrive on the installation. If needed, the QI will coordinate with local community government agencies to request support. QI will comply with incident reporting requirements through IOC.

1.8.4 INCIDENT COMMANDER

The IC will, with the assistance of the IOSC, maintain accountability of response personnel and equipment in the field, record response actions taken, and provide updates to the EOC as the situation progresses, keeping them apprised of any progress and any developing threats or challenges.

a. The on-site FES Incident Commander will remain in command until the IRT has secured the site and all immediate threats to human health and safety have been mitigated. When the FES mission is complete, the Fire Chief will transfer the IC responsibilities to the IOSC. FES will continue to provide support for as long as they are needed on the site.

b. The IOSC will accept the duties of the IC, request free product recovery or other support required, and monitor conditions to ensure containment is not breached until spilling organization begins cleanup activities or OSRO support arrives.

1.8.5 THE EMERGENCY OPERATION CENTER (EOC)

The Fort Cavazos EOC is located in Room W217 of Building 1001. The Garrison Commander or designated representative will exercise command authority of all assigned and attached military and civilian personnel and resources, whether they are on or off the installation during the disaster response and recovery.

1.9 FACILITY DIAGRAMS

Fort Cavazos encompassing 218,823 acres or 342 square miles of Bell and Coryell Counties. The installation includes approximately 136,094 acres of maneuver training area, 64,000 acres as a live fire impact area, and 22,026 acres for the installation’s cantonment areas. In addition to the vast training area, the installation has four developed geographical areas as depicted in Diagram 1 below: Main Cantonment (MC), West Fort Cavazos (WFC), North Fort Cavazos (NFC), and Belton Lake Outdoor Recreational Area (BLORA). The SPCCP contains diagrams that focus on the installation as a whole and incorporates the use of the Fort Cavazos Phantom Warrior WebMap, to meet the mapping requirements for an installation of this size.

a. For spill response purposes, the diagrams in this plan focus on capitalized and commercial bulk fuel storage facilities with a potential for a medium to worse-case scenario spill.

b. Probable cause for a medium or worse-case scenario spill at any of these locations is: Rupture of tanks; failure of pumps, pipes, transfer lines; or incidents during loading/unloading operations.

1.9.1 FORT CAVAZOS STORMWATER DRAINAGE

Any spill occurring in one of the developed geographical areas, from a maintenance, oil transfer, or bulk fuel facility; storage tank; or oil filled equipment, that cannot be contained on site, will be channeled through the Fort Cavazos Municipal Separate Storm Sewer System (MS4) before entering natural surface stormwater drainage features such as unnamed tributaries, seasonally-dry creek beds, and rivers that eventually discharge outside of the Fort Cavazos boundaries and into South Nolan Creek, Belton Lake, or Stillhouse Lake. Diagram 1 provides an overview of the primary stormwater pathways that convey stormwater drainage from developed geographic areas, to major waterbodies.

1.9.2 DIAGRAM REQUIREMENTS

a. A diagram has been developed for each facility identified in Section 4 with the potential for a medium to worse-case scenario spill. The diagrams include the following items as required by 30 TAC §112.7(a)(3):

- Aboveground storage tanks (to include generator belly tanks)
- Underground storage tanks
- Storage area(s) where mobile or portable containers are located
- Transfer stations such as oil transfer areas including loading/unloading racks and loading/unloading areas
- Oil-filled equipment such as hydraulic operating systems or manufacturing equipment (including elevators and fire suppression system sumps)
- Oil-filled electrical transformers
- Connecting piping (see Section 1.9.3b.)
- Emergency evacuation routes from gated areas
- Spill path of discharge
- Discharge drainage controls

b. Due to the size of the installation and quantity of items required on each map, the DPW GIS office has created an SPCCP map layer for the Fort Cavazos Phantom Warrior WebMap. This allows any agencies on Fort Cavazos to view or "zoom" to a specific location on the interactive map (i.e., spill site location), apply the layer filters, and identify any SPCCP relevant details at that location. The scale of the diagrams included in this plan does not allow for all of the connecting piping to be annotated; however, they are identified in the to the SPCCP map layer for the Fort Cavazos Phantom Warrior interactive map.

1.9.3 BULK FUEL STORAGE AREA DIAGRAMS

Below is the list of bulk fuel storage area diagrams, which includes brief description of the stormwater drainage route for each location.

a. Diagram 2 Main Bulk Fuel Supply Facility (BFSF) located at 4614 Logistics Avenue.

b. Diagram 3 Robert Gray Alert Farm (RGAF) located at 4642 S. Clarke Road.

a. Diagram 4 Yoakum-Defrenn Army Heliport (YDAHP) located at 3675 Warrior Way.

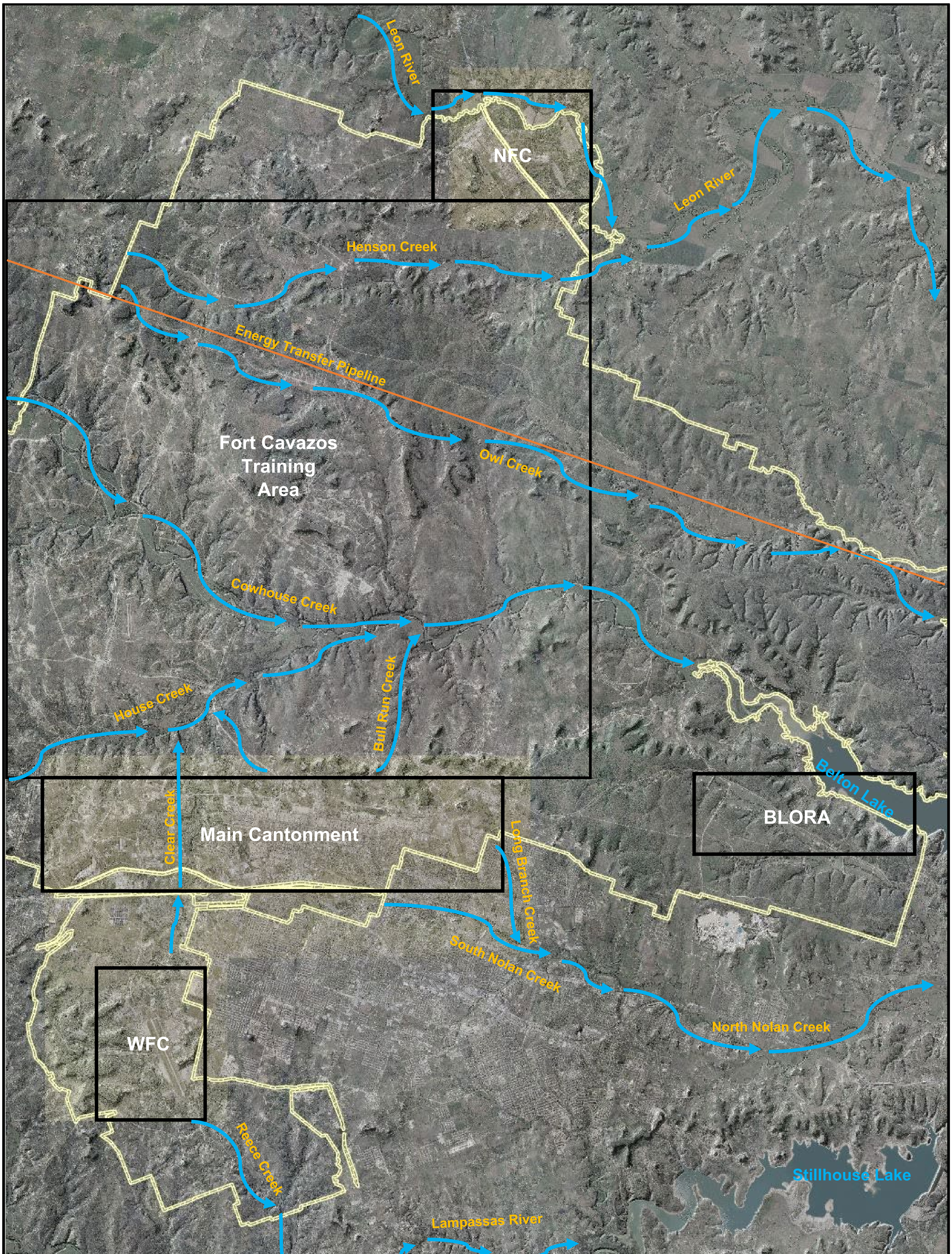
b. Diagram 5 AAFES Clear Creek Express located at 2331 Stadium Way.

c. Diagram 6 AAFES Commanche Express located at 3433 W. Tank Destroyer Boulevard.

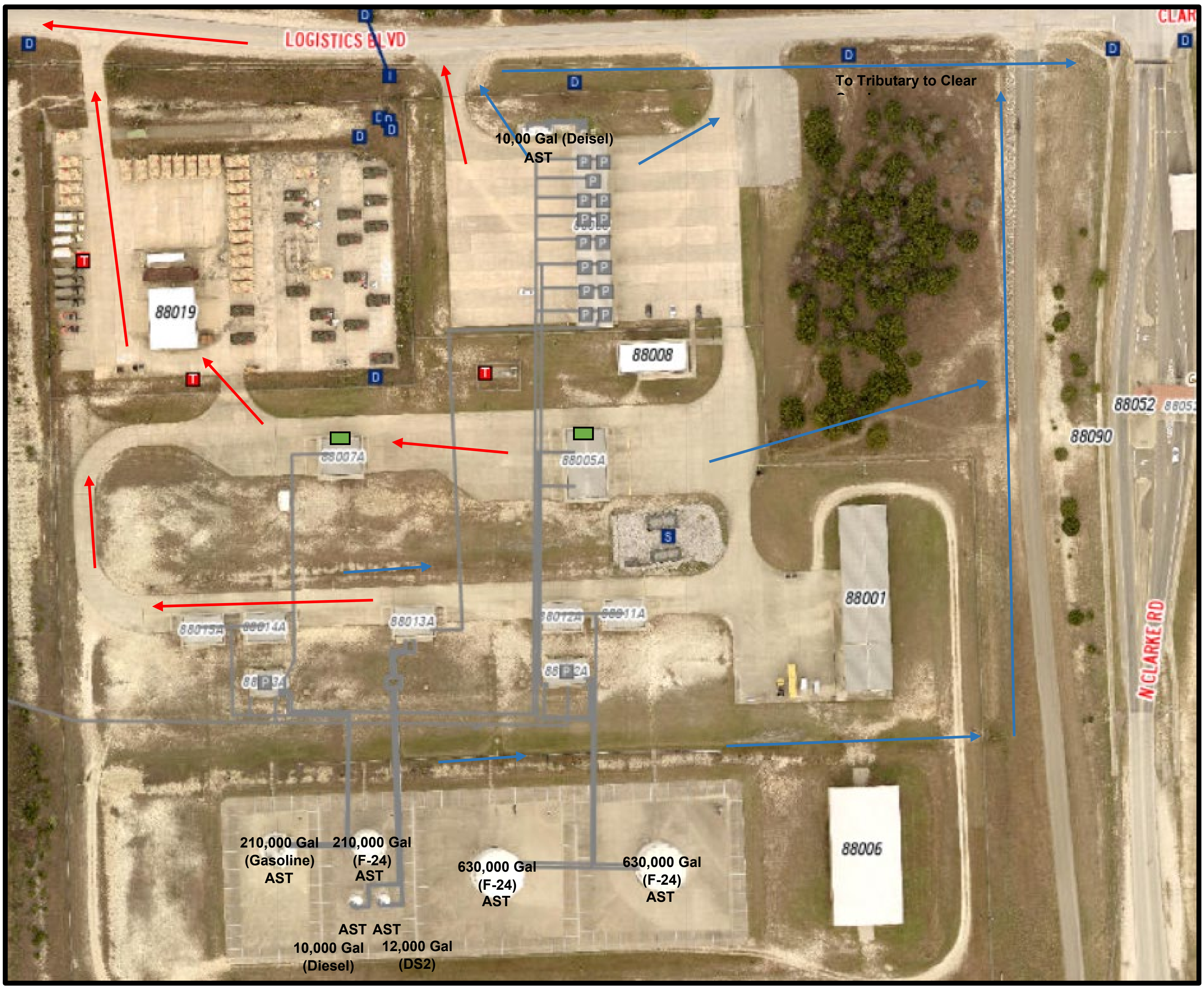
d. Diagram 7 AAFES WFC Express located at 302 S. Clarke Road.

FRP Diagram 1

Developed Geographic Area and Stormwater Drainage Routes



FRP Diagram 2
Main Cantonment Bulk Fuel Supply Facility

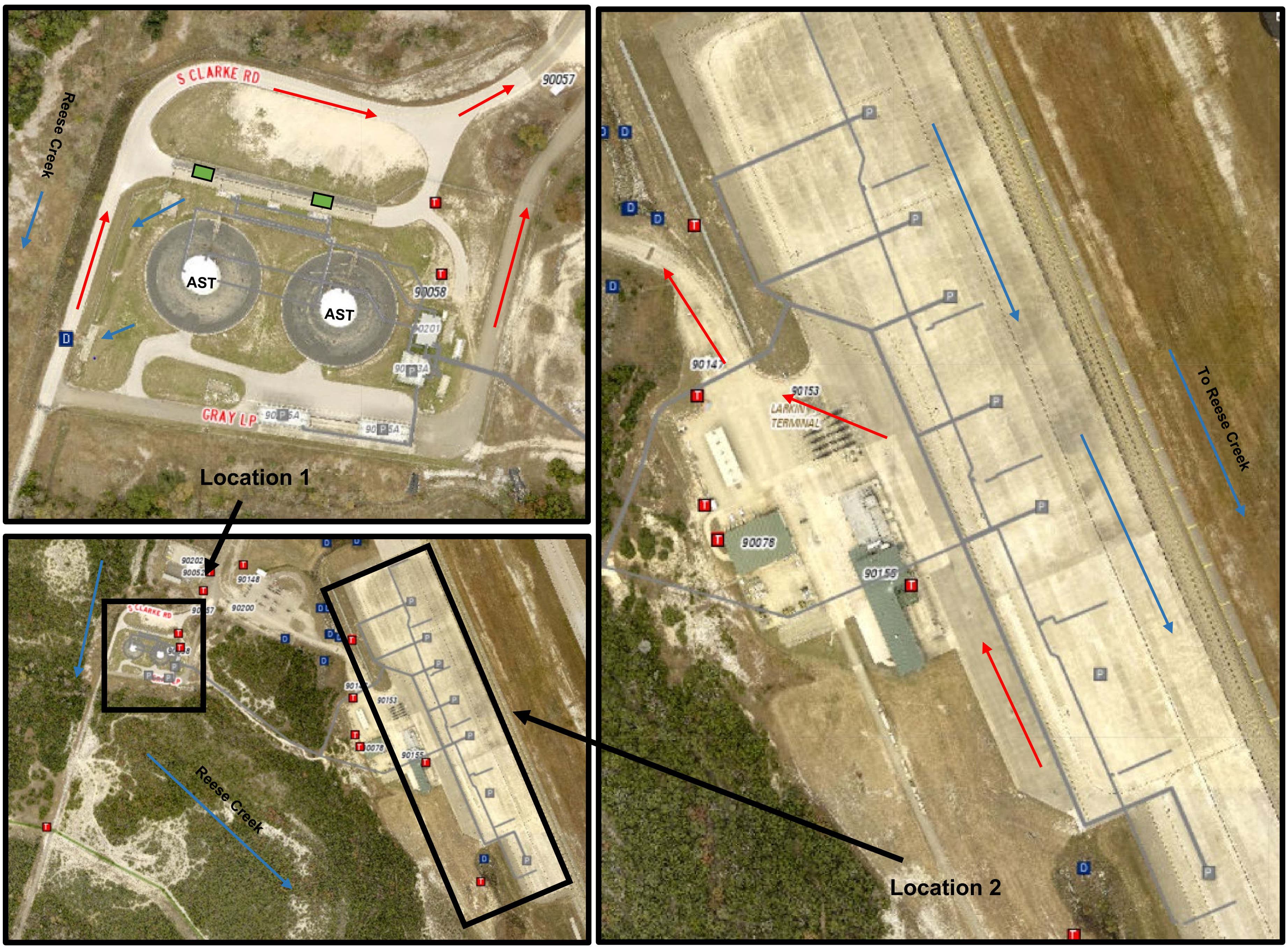


A spill occurring in this area will be channeled through the Fort Cavazos MS4 in a westerly direction to a containment pond. Discharge from the containment pond will continue westerly via an unnamed tributary to Clear Creek, to House Creek, to Cowhouse Creek, and on to Belton Lake. This site contains:

- Seven ASTs of varying sizes (Tank size and content on diagram)
- Two fuel transfer stations
- Over 5000 feet of connecting piping
- 15 Pump Stations
- Three oil filled electrical transformers

T	Oil Filled Electrical Transformers
D	Stormwater Discharge Point
P	Fuel Pump Point
	Stormwater Directional Flow
	Evacuation Route
	Connecting Piping
	Fuel Transfer Stations

FRP Diagram 3
Robert Gray Alert Farm



A spill occurring in this area will be channeled in a southeasterly direction through cement and vegetated ditches and unnamed tributaries to Reese Creek, southerly to the Lampassas River and then easterly to Stillhouse Lake. Probable Cause: Rupture of tanks; failure of pumps, pipes, transfer lines; or incidents during loading/unloading operations. This site is split into two areas with over 10,000 feet of connecting piping.

Location 1 – Tank Farm contains:

- Two 541,280 Gallon ASTs containing F-24
- Two fuel transfer station
- Two oil filled electrical transformers

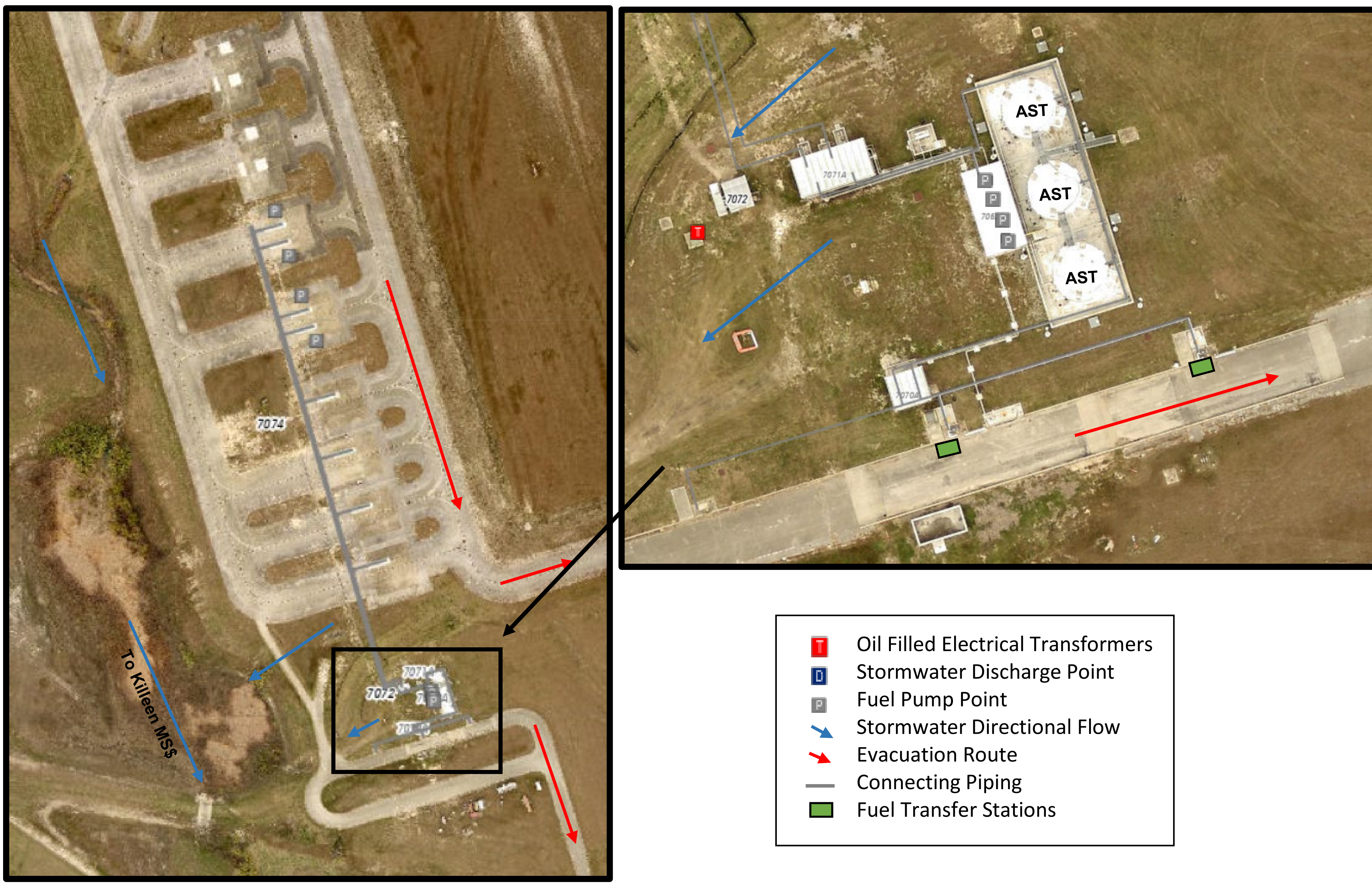
Location 2 - Refuel Point contains:

- Seven pump stations (hydrants)
- Five oil filled electrical transformers (in close proximity)

T Oil Filled Electrical Transformers
D Stormwater Discharge Point
P Fuel Pump Point
Stormwater Directional Flow
Evacuation Route
Connecting Piping
Fuel Transfer Stations

FRP Diagram 4

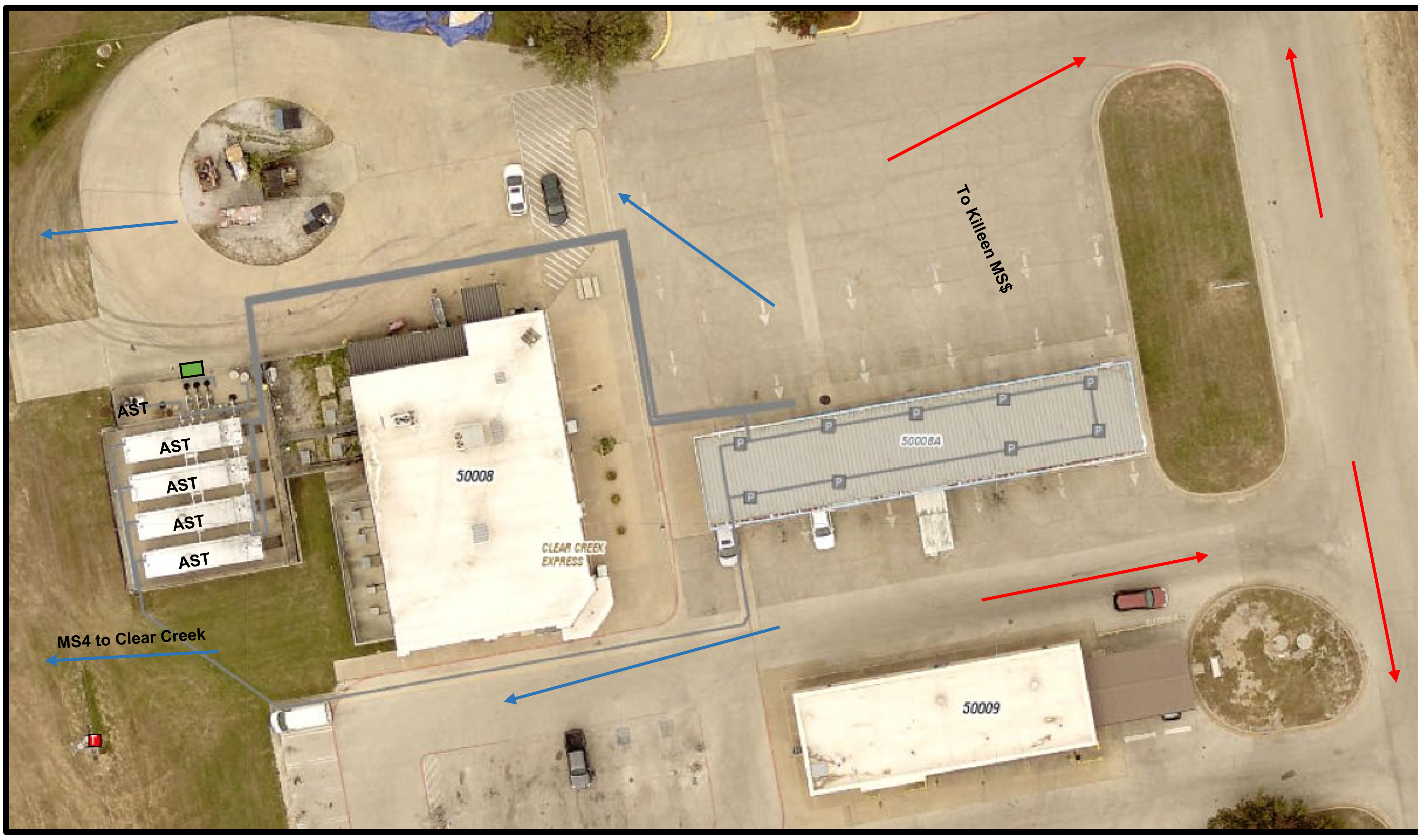
Yoakum-Defrenn Army Heliport Rapid Refuel



A spill occurring in this area will travel south westerly through the Fort Cavazos MS4 and discharge into Airfield Lake, north to the City of Killeen MS4, to South Nolan Creek, east to North Nolan Creek, and to Belton Lake. Probable Cause: Rupture of tanks; failure of pumps, pipes, transfer lines; or incidents during loading/unloading operations. This site contains:

- Three 50,000 Gallon ASTs containing F-24
- Over 3000 feet of connecting piping
- Two fuel transfer station
- One oil filled electrical transformer
- Eight pump stations (hydrants)

FRP Diagram 5
AAFES Clear Creek Express



A spill occurring in this area will travel south westerly through the Fort Cavazos MS4 and discharge into Airfield Lake, north to the City of Killeen MS4, to South Nolan Creek, east to North Nolan Creek, and to Belton Lake. Probable Cause: Rupture of tanks; failure of pumps, pipes, transfer lines; or incidents during loading/unloading operations. This site contains:

- Four 12,000 Gallon ASTs containing gasoline
- One 280 Gallon AST containing Off-Spec Fuel
- Over 1,800 feet of connecting piping
- Onw fuel transfer station
- One oil filled electrical transformer
- Nine pump stations (hydrants)

T Oil Filled Electrical Transformers
D Stormwater Discharge Point
P Fuel Pump Point
Stormwater Directional Flow
Evacuation Route
Connecting Piping
Fuel Transfer Stations

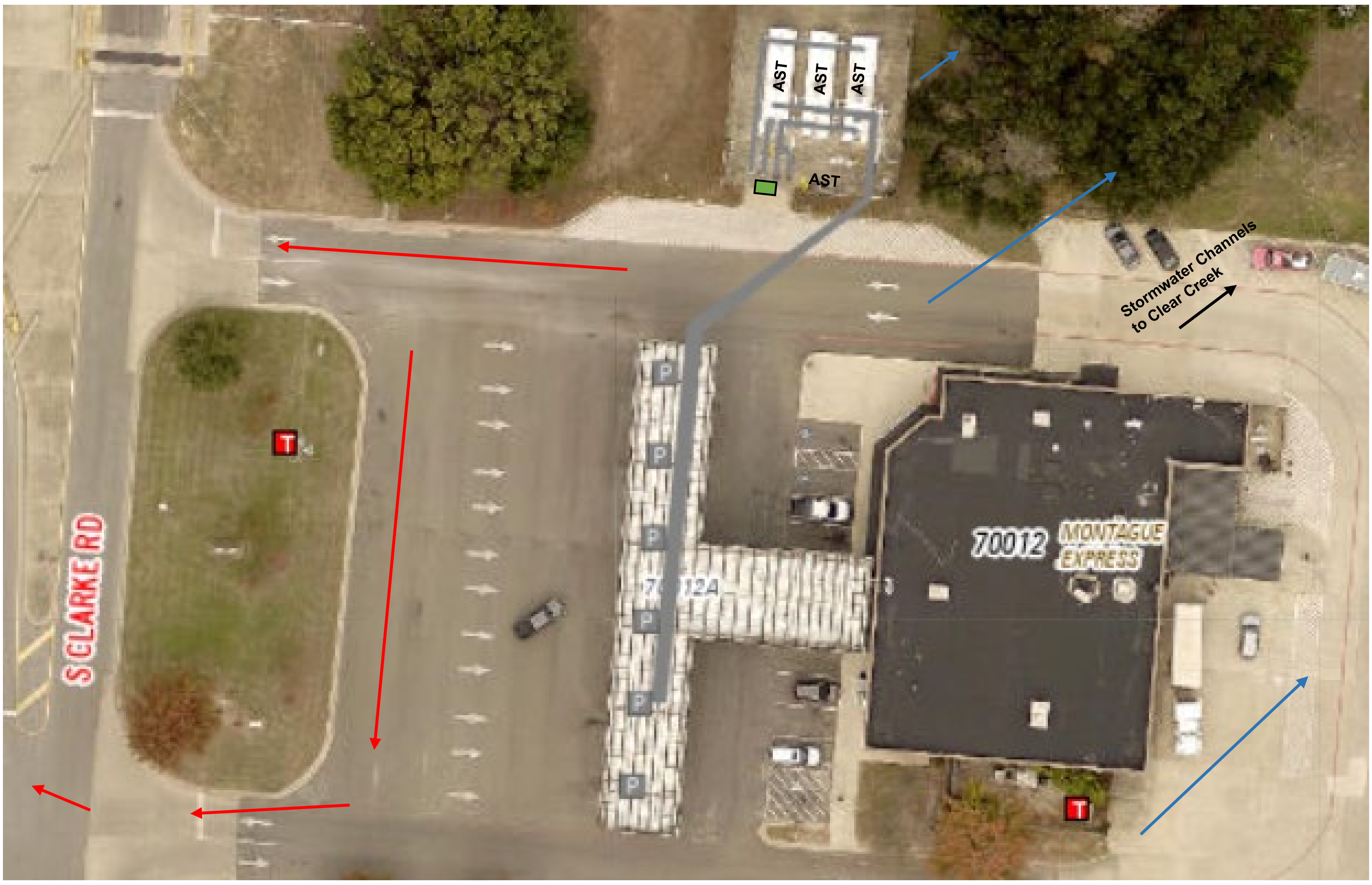
FRP Diagram 6
AAFES Commanche Express

A spill occurring in this area will be channeled through the Fort Cavazos MS4 in a westerly direction to Clear Creek, to House Creek, to Cowhouse Creek, and on to Belton Lake. Probable Cause: Rupture of tanks; failure of pumps, pipes, transfer lines; or incidents during loading/unloading operations. This site contains:

- Three 12,000 Gallon ASTs containing gasoline,
- One 280 Gallon AST containing off-spec fuel
- One fuel transfer station
- Approximately 1000 feet of connecting piping
- Eight Pump Stations
- Three oil filled electrical transformers

T Oil Filled Electrical Transformers
D Stormwater Discharge Point
P Fuel Pump Point
Stormwater Directional Flow
Evacuation Route
Connecting Piping
Fuel Transfer Stations

FRP Diagram 7
AAFES WFC (Montegue) Express



A spill occurring in this area will travel through stormwater drainage channels in a northwesterly direction to a drainage culvert passing under I-14 and then in a northerly direction to a containment pond. Discharge from the containment pond will continue westerly via an unnamed tributary to Clear Creek, to House Creek, to Cowhouse Creek, and on to Belton Lake. This site contains:

- Three 12,000 Gallon ASTs containing gasoline
- One 280 Gallon AST containing off-spec fuel
- One fuel transfer station
- Over 2000 feet of connecting piping
- Six Pump Stations
- Two oil filled electrical transformers

T Oil Filled Electrical Transformers
D Stormwater Discharge Point
P Fuel Pump Point
Stormwater Directional Flow
Evacuation Route
Connecting Piping
Fuel Transfer Stations

AST
UST
Connecting Piping
POL Storage Area
Transfer Station
Oil Filled Equipment
Oil Filled Electrical Transformers
Evacuation Route
Stormwater Discharge Points
Spill Discharge Drainage Path

2. FACILITY INFORMATION

Installation Name: III Armored Corps and Fort Cavazos

Location: Headquarters, III Armored Corps and Fort Cavazos
Building 1001, 761st Tank Battalion
Fort Cavazos, TX 76544-5028

County: Bell and Coryell

Latitude:	31	Degrees	07	Minutes	51	Seconds
Longitude:	97	Degrees	46	Minutes	04	Seconds

Phone Number: (254) 287-1110

Wellhead Protection Area: Not Applicable

Owner/Operator: Department of the Army

Owner/Operator: Location (Street Address): Same as above

Qualified Individual(s): Timi Dutchuk/TBD

Date of Oil Storage Start-up: 1942

Current Operations: Army military training reservation

Date(s) and Type(s) of Substantial Expansion(s): Ongoing since 1942

2.1 QI DUTIES AND RESPONSIBILITIES

DPW ENV Chief, serves as the QI for Fort Cavazos. The Environmental Management Branch Chief, serves as the alternate. Contact information is provided in Section 1.1, Qualified Individuals. The following information is a summary of the QI's responsibilities, in coordination with the FES and pre-existing support, in the event of emergencies:

Note: QI/Alternate QI contact information is located in Section 1.1, QI/Alternate QI Information.

2.1.1 REGULATORY DUTIES

The regulatory duties [40 CFR 120(h)(3)(ix)] of the QI are listed below:

Note: Due to the size and complexity of the Fort Cavazos Military Installation, some of the regulatory responsibilities of the QI are sub tasked or conducted by other listed organizations. For this reason, the QI will work in conjunction with the designated entity to ensure all requirements are met. The designated entity is listed next to the requirement.

a. Activate alarms from the loading rack, manifold, firehouse, warehouse, or office to notify all base personnel – FES

b. Notify all response personnel per the Emergency Response Notification Phone List – QI/Alternate QI

c. Identify character, exact source, amount, and extent of release, as well as other items needed for notification – FES

d. Notify and provide necessary information to the appropriate federal, state, and local authorities with designated response roles, including the NRC, SERC, and LEPC – QI/Alternate QI

e. Using knowledge of the HAZMAT on-site, assess interaction of spilled substance with water and/or other substances stored at the facility and notify response personnel at the scene of that assessment – FES

f. Assess possible hazards to human health and environment due to the release. The assessment will consider both direct and indirect effects of the release (gases released, effects of any hazardous surface water runoffs from water or chemical agents used to control fire and explosion). Considerations will also be given to environmental hazards posed by releases from the foam suppression systems at the facility - FES

g. Assess and implement prompt isolation actions to control, contain, and remove substance released – FES

h. Use authority to immediately access funding to initiate cleanup activities. The QI is granted full authority to implement corrective actions – DRM/MICC

i. Direct cleanup activities until properly relieved of this responsibility – QI/Alternate QI

2.1.2 QI AUTHORITY AND RESPONSIBILITY

The QI has initial command and control responsibilities, in coordination with the IC for any oil or HAZMAT spill addressed by this plan. As a minimum, the QI and designated alternates must have:

a. Authority to activate and contract with oil spill removal organizations.

b. Authority to act as liaison with the federal OSC.

c. Authority to obligate appropriate funding to support all necessary or directed oil response activities. However, government contracts do not allow the QI to obligate funds for spill cleanup, only the Contracting Officers can do that. The QI will verify the necessity to activate a spill contract and request that the Contracting Officer execute the contract.

2.2 SELF-INSPECTION, DRILL/EXERCISE AND RESPONSE TRAINING

The self-inspection, drill/exercise, and response training and exercise plan for Fort Cavazos has been developed to meet the requirements of 40 CFR 112.21, to include: A checklist and record of inspections for tanks, secondary containment, and response equipment; a description of the drill/exercise program; a description of the training program; logs of training sessions and drills/exercises.

2.2.1 SELF-INSPECTIONS

Self-inspection and maintenance procedures and schedules for response equipment have been developed in accordance with equipment manufacturer guidance and are conducted, updated, and maintained by the owning/operating response organization and will be made available upon request.

2.2.2 DRILL/EXERCISE PROGRAM

The purpose of the equipment deployment exercises is to ensure that response equipment is appropriate for the operating environment in which it is intended to be used and that operating personnel are trained in its deployment and operation.

a. This program includes the following components.

- Personnel who would normally operate or supervise the operation of the response equipment participate in the exercise.
- Personnel demonstrate the ability to deploy and operate the equipment, while wearing appropriate personal protective equipment.
- Operating personnel participate in exercises or responses on an annual basis in order to ensure that they remain trained and qualified to operate equipment in the operating environment.
- Response equipment is in good operating condition.
- Equipment is appropriate for the intended operating environment.
- Equipment is be operated during the exercise.
- There is a maintenance program for all response equipment.

NOTE: It is not necessary to deploy every piece of each type of equipment, as long as all equipment is included in a periodic inspection and maintenance program intended to ensure that the equipment remains in good working order.

b. Announced Exercises - Fort Cavazos response personnel participate in at least one large scale installation-wide spill response exercise each year that meets the OPA 90 requirements for Equipment Deployment Exercises.

c. Unannounced Exercises – The annual unannounced exercise requirement is necessary to maintain the level of preparedness to respond to a spill effectively. Fort Cavazos response personnel participate in at least one unannounced spill response exercise each year that meets the OPA 90 requirements. A response to an actual spill may be considered as an unannounced exercise, if the response is self-evaluated and required exercise objectives were met and documented.

d. DPW Environmental Personnel maintains a spill response log of all incidental and reportable spills, with corresponding electronic files on each event. All exercises will be logged in the spill response log with supporting documents.

2.2.3 RESPONSE TRAINING

Personnel who respond to or may encounter emergency hazardous substance releases must complete HAZWOPER training. Initial training and annual refresher training is required for all levels of responders—from awareness level to the Incident Commander. A record of the training or competency is maintained by each response organization. If a statement of competency is made, the employer shall keep a record of the methodology used to demonstrate competency. Response Training for Fort Cavazos personnel is based on the duties and function to be performed by each response organization in accordance with 29 CFR 1910.120(q)(6) as follows:

a. First responder awareness level. First responders at the awareness level are individuals who are likely to witness or discover a hazardous substance release and who have been trained to initiate an emergency response sequence by notifying the proper authorities of the release. First responder training at the awareness level is provided by the DPW Environmental Division through the Environmental Compliance Officer Course and provides sufficient training or experience to objectively demonstrate competency in the following areas:

- An understanding of what hazardous substances are, and the risks associated with them in an incident.
- An understanding of the potential outcomes associated with an emergency created when hazardous substances are present.
- The ability to recognize the presence of hazardous substances in an emergency.
- The ability to identify the hazardous substances, if possible.
- An understanding of the role of the first responder awareness individual in the employer's emergency response plan including site security.
- The ability to realize the need for additional resources, and to make appropriate notifications to the command or leadership.

b. First responder operations level. First responders at the operations level are individuals who respond to releases or potential releases of hazardous substances as part of the initial response to the site for the purpose of protecting nearby persons, property, or the environment from the effects of the release. They are trained to respond in a defensive fashion without actually trying to stop the release. Their function is to contain the release from a safe distance, keep it from spreading, and prevent exposures. DPW Environmental Response Team personnel who meet this category criteria, receive 24hr initial training from certified HAZWOPER Trainers to include awareness level training as well as at least eight hours of operations level training or have had sufficient experience to objectively demonstrate competency in the following areas in addition to the awareness level of training:

- Knowledge of the basic hazard and risk assessment techniques.
- Know how to select and use proper personal protective equipment provided to the first responder operational level.
- An understanding of basic hazardous materials terms.
- Know how to perform basic control, containment and/or confinement operations within the capabilities of the resources and personal protective equipment available with their unit.
- Know how to implement basic decontamination procedures.
- An understanding of the relevant standard operating procedures and termination procedures.

c. Hazardous materials technician. Hazardous materials technicians are individuals who respond to releases or potential releases for the purpose of stopping the release. They assume a more aggressive role than a first responder at the operations level in that they will approach the point of release in order to plug, patch or otherwise stop the release of a hazardous substance. FES personnel received 40hrs of certified HAZWOPER initial training equal to the first responder operations level and competency in the following area:

- Know how to implement the employer's emergency response plan.
- Know the classification, identification and verification of known and unknown materials by using field survey instruments and equipment.
- Be able to function within an assigned role in the Incident Command System.
- Know how to select and use proper specialized chemical personal protective equipment provided to the hazardous materials technician.
- Understand hazard and risk assessment techniques.
- Be able to perform specialized control, containment, and/or confinement operations within the capabilities of the resources and personal protective equipment available.
- Understand and implement decontamination procedures.
- Understand termination procedures.
- Understand basic chemical and toxicological terminology and behavior.

d. On scene incident commander. Incident commanders, who will assume control of the incident scene beyond the first responder awareness level, has receive at least 24 hours of training equal to the first responder operations level and has competency in the following areas:

- Know and be able to implement the employer's incident command system.
- Know how to implement the employer's emergency response plan.
- Know and understand the hazards and risks associated with employees working in chemical protective clothing.
- Know how to implement the local emergency response plan.
- Know of the state emergency response plan and of the Federal Regional Response Team.
- Know and understand the importance of decontamination procedures.

e. Incident Command System Training. The National Incident Management System (NIMS) Training Program identifies those courses critical to train personnel capable of implementing all functions of emergency management. This program establishes the NIMS core curriculum to ensure it adequately trains emergency and incident response personnel to all concepts and principles of each NIMS component. The following baseline courses meet the requirements for specific functions of the IOSC.

- IS-700B Introduction to NIMS: This course introduces the NIMS concept and provides a consistent nationwide template to enable all government, private-sector, and nongovernmental organizations to work together during domestic incidents.
- ICS-100 Introduction to the Incident Command System: This course introduces ICS and provides the foundation for higher level ICS training. It describes the history, features and principles, and organizational structure of the system.
- Position-Specific Training: These courses are designed to provide State and local-level emergency responders with a robust understanding of the duties, responsibilities, and capabilities of Command and General Staff members. Position-specific training course required for IOSC/IC - E/L 950: NIMS ICS All-Hazards Position Specific Incident Commander.

f. Trainers. Trainers who teach any of the above training subjects shall have satisfactorily completed a training course for teaching the subjects they are expected to teach or have the training and/or academic credentials and instructional experience necessary to demonstrate competent instructional skills and a good command of the subject matter of the courses they are to teach. The FES has personnel certified to conduct the 8hr HAZWOPER Annual Refresher Course. The DPW Environmental Trainer is qualified to provide first responder awareness level training to appointed Environmental Compliance Officers who provide reporting and response awareness to Unit/Activity personnel.

g. Refresher training. Those employees who are trained in accordance with 29 CFR 1910.120(q)(6) receive annual refresher training of sufficient content and duration to maintain their competencies or demonstrate competency in those areas at least yearly.

3. HAZARD EVALUATION

Hazard identification and evaluation assists in planning for potential discharges. This section examines the oil storage facilities at Fort Cavazos to determine where discharges are likely to occur and the potential impact.

3.1 HAZARD IDENTIFICATION

Fort Cavazos has over 600 fuel storage tanks across the installation. An overview of the POL storage and transfer locations on the installation, to include spill pathways from each area, is provided in Section 1.9, Facility Diagrams. A complete inventory of all POL tanks managed by Fort Cavazos, including type/size of container, location, and product contained, is maintained by DPW ENV.

3.1.1 DEFENSE FUEL STORAGE POINT (DFSP)

The DFSP at Fort Cavazos is a government-owned contractor-operated facility which stores and supplies bulk fuel to military units at Fort Cavazos. The DFSP contractor operates under a DLA contract. The DFSP contractor is responsible for the operations, routine inspections, and maintenance of the DFSP. DFSP consists of three separate fuel storage and transfer operations within the confines of the military installation: the Robert Gray Alert Facility (RGAF), the YDAHP RRF, and the Main Cantonment BFSF.

a. Facility Information:

1) The Main Cantonment BFSF is the main bulk fuel storage and distribution facility for F-24, MOGAS, diesel, and E-85 at Fort Cavazos. Commercial tank trucks deliver fuels to this facility at the facility’s unload stands. The BFSF distributes fuel in bulk to other installation fuel use sites and users by tank trucks or military refueling trucks using load racks and load stands. The BFSF also dispenses fuel to military vehicles at a retail dispensing facility. The bulk storage ASTs at the DSFP facilities are field-constructed atmospheric steel tanks built to either American Petroleum Institute (API) 12-C or API 650 standards.

2) The RGAAF consists of a bulk F-24 storage facility that supplies F-24 to a hydrant at the South Ramp at the RGAAF. The hydrant refuels fixed wing aircrafts and loads military refueling trucks. This site also includes a parking apron for parking refuelers and hydrant hose trucks. Commercial and DFSP tank trucks deliver F-24 to the RGAAF at the facility unload stand.

3) The Yoakum-Defrenn Army Heliport (YDAHP) Rapid Refuel Facility (RRF) consists of a bulk F-24 tank farm that supplies F-24 to a hydrant at the YDAHP. The hydrant refuels rotary wing aircrafts and loads military refueling. Commercial and DFSP tank trucks deliver F-24 to the YDAHP at the facility unload stand.

b. Discharge Hazards:

1) Main Cantonment BFSF: Rupture of tanks or pipes may cause reportable spills. Failure of pumps, pipes, transfer lines, or other transfer equipment also may cause a reportable spill. Loading/ unloading operations may result in a maximum spill of 8,000 gallons.

2) Refueling at YDAHP: Failure of re-supply tankers and loading/unloading operations may result in a maximum spill of 8,000 gallons. Rupture of tanks or pipes may cause reportable spills. Failure of pumps, pipes, transfer lines, or other transfer equipment also may cause a reportable spill.

3) RGAAF: Rupture of tanks or pipes may cause reportable spills. Failure of pumps, pipes, transfer lines, or other transfer equipment also may cause a reportable spill. Loading/unloading operations may result in a maximum spill of 8,000 gallons.

3.1.2 FOOD PREPARATION FACILITIES

Food preparation facilities at Fort Cavazos that use grease containers to store used cooking oil and grease are the installation dining facilities and the AAFES operated food establishments. The two organizations have two separate cooking oil vendors that supply and service their used cooking oil containers. Both vendors provide 280-gallon steel outdoor containers.

3.1.3 MAINTENANCE FACILITIES

The majority of oil storage areas on the installation are located in equipment, vehicle, or aircraft maintenance facilities on the installation. This includes military wheeled/track and aircraft maintenance facilities, and DoD civilian or contractor operated maintenance facilities.

a. Type Storage Containers:

1) The ASTs storing oil at Fort Cavazos are typically found in maintenance facilities and are primarily shop-fabricated UL 142 double-walled steel tanks, or primary steel wall, high-density polyethylene (HDPE)-lined, encased in concrete tanks. These shop-fabricated ASTs range in size from 125 to 20,000 gallons, the most common size is 500 gallons.

2) The maintenance facilities also store 55-gallon steel drums of POL and HAZMAT in adequate secondary containment by using secondary containment units (SCUs), spill pallets, and HAZMAT sheds, with integrated secondary containment.

3) Mobile Fuel Tankers (MFTs) or Mobile Fuel Systems (MFSs) are stored at the military maintenance facilities.

b. Discharge Hazard:

1) Motorpools and Hangars: Lack of maintenance, damage, or misuse of oil-water separators can result in reportable oil spill. Lack of maintenance, damage, misuse, or explosion of refuelers could result in a spill.

2) Mobile and Portable Fueling Equipment: Ruptured tanks or failure of transfer connections could spill up to 5,000 gallons. Transit accidents, lack of maintenance, damage, misuse, or explosion of a refueler could result in a spill of up to 3,000 gallons.

3) DPW Maintenance Facility: Rupture of tanks or pipes may cause reportable spills. Failure of pumps, pipes, transfer lines, or other transfer equipment also may cause a reportable spill. Loading/unloading operations may result in a maximum spill of 5,000 gallons.

4) AFSBn, Consolidated Shop: Rupture of tanks or pipes may cause reportable spills. Loading/unloading operations may result in a maximum spill of 5,000 gallons.

5) AFSBn, Component Overhaul: Rupture of tanks or pipes may cause reportable spills. Loading/unloading operations may result in a maximum spill of 5,000 gallons.

6) AFSBn, Used oil bunker near Building 88036: Tank rupture may result in a reportable maximum spill of 1,000 gallons.

3.1.4 OIL TRANSFER OPERATIONS

In addition to the DFSP bulk fuel transfer operations, Fort Cavazos established end-use oil transfer operations. These transfer operations consist primarily of tank truck oil transfer and/or retail refueling. Below are the locations of these facilities and their associated hazards.

a. Recycle Storage Facility: Failure of tankers and loading/unloading operations may result in a maximum spill of 5,000 gallons. Rupture of tanks or pipes may cause reportable spills. Failure of pumps, pipes, transfer lines, or other transfer equipment also may cause a reportable spill.

b. Active Sanitary Landfill: Rupture of tanks or pipes may cause reportable spills. Loading/unloading operations may result in a maximum spill of 3,000 gallons.

c. Post Exchange Service Centers (AAFES): Rupture of tanks or pipes may cause reportable spills. Failure of pumps, pipes, transfer lines, or other transfer equipment also may cause a reportable spill. Loading/unloading operations may result in a maximum spill of 5,000 gallons.

d. BLORA: Loading and unloading operations may result in a maximum spill of 3,000 gallons.

e. North Fort Cavazos: Rupture of tanks or pipes may cause reportable spills. Failure of pumps, pipes, transfer lines, or other transfer equipment also may cause a reportable spill. Loading/unloading operations may result in a maximum spill of 5,000 gallons.

3.2 VULNERABILITY ANALYSIS

This section addresses the potential effects of a spill from the various oil storage facilities and the vulnerability of sensitive environments and public facilities on Fort Cavazos.

3.2.1 SPILL PATHWAYS

Fort Cavazos consists of extensive training areas and four developed geographical areas; the main cantonment area, WFC, BLORA, and NFC. Probable oil spill pathways for the training areas and each geographical areas are described in Section 1.9.

3.2.2 STORMWATER DRAINAGE

Stormwater Runoff from the Fort Cavazos Main Cantonment and WFC is channeled through the Fort Cavazos MS4, which consists of a series cement or vegetation lined ditches, stormwater inlets/outlets, and pipes. Depending on the location on the installation, stormwater exits Fort Cavazos via Cowhouse Creek, Reese Creek, Longbranch Creek, Killeen MS4, or the FCFH MS4. Stormwater from NFC and BLORA travel through a similar unregulated drainage system, to the Leon River and Belton Lake. Stormwater runoff on Fort Cavazos is regulated by the TCEQ through TPDES stormwater and wastewater permits. The permits require Fort Cavazos to prepare and implement stormwater pollution prevention and management plans that include utilizing best management practice, routing inspections, and monitoring/sampling to ensure that Fort Cavazos does not discharge pollutants to the environment. TCEQ has issued Fort Cavazos authorization to discharge under the following TPDES Permits:

a. TPDES General Permit TXR 040000, for small MS4s – This permit regulates municipal the Fort Cavazos MS4 which required pollution prevention measures for stormwater discharge in the Main Cantonment and WFC geographical areas.

b. TPDES General Permit TRX 050000 a multisector general permit for industrial stormwater – This permit regulates stormwater discharge from identified industrial facilities on the installation. Among those are oil storage or transfer areas located at P2 Services, CU, Fort Cavazos Recycle Facility, RGAAF and YDAHP.

c. TPDES Permit No. WQ0002233000, for the Fort Cavazos Vehicle Wash Facility – This permit regulates the stormwater discharged from the vehicle maintenance facilities located along Ivy Division Road, Landfill Lake, Birdbath Lake, and East Lake.

3.2.3 STORMWATER SYSTEM PIPING

The piping which makes-up both the regulated and unregulated stormwater systems vary in size and type. The smallest pipe size in the system consists of three-inch roof drains; the largest pipe in the system is 60-inch reinforced concrete piping, located under Warrior Way, south of YDAHP. The types of pipes utilized on-post include reinforced concrete, corrugated metal, polyvinyl chloride, and ductile iron. The drainage conditions on-post are acceptable, with minimal ponding or flooding.

3.2.4 OIL WATER SEPARATORS (OWS)

Fuel storage and transfer operations (i.e., storage tank dikes, aviation fuel loading and unloading areas, and the MOGAS/diesel fueling areas) are designed to divert stormwater runoff which may be contaminated, through OWSs, prior to discharging into the Fort Cavazos MS4. These safeguards have been installed to protect the environment and ensure compliance with the conditions of TPDES discharge authorizations issued to Fort Cavazos.

3.3 POTENTIAL IMPACT TO THE ENVIRONMENT

Potential impacts to the environment were evaluated for vulnerable waterways, threatened or endangered species, and community impact.

3.3.1 VULNERABLE WATERWAYS

Vulnerable waterways include: Clear Creek, Bull Run Creek, House Creek, Cowhouse Creek, Long Branch Creek, South Nolan Creek, Nolan Creek, Reese Creek, Leon River; Lampassas River, Stillhouse Hollow Lake, and Belton Lake.

3.3.2 THREATENED OR ENDANGERED SPECIES

Any vegetation, wildlife or fish in the spill pathway would be at risk; however, the Federally-listed threatened and endangered species identified for Bell and Coryell counties which are potentially at risk from a spill include the following bird species: golden-cheeked warbler (*Dendroica chrysoparia*), arctic peregrine falcon (*Falco peregrinus anatum*), whooping crane (*Grus americana*), and piping plover (*Charadrius melodus*).

3.3.3 POTENTIAL COMMUNITY IMPACTS

The spill pathways from some of the Fort Cavazos storage facilities pass ½ mile or less from residential areas, schools, or childcare facilities, which could be affected by fumes or potential for fire hazards. Additionally, a spill reaching Belton Lake and Stillhouse Hollow Lake could limit the recreational activities.

3.4 ANALYSIS OF THE POTENTIAL FOR AN OIL SPILL

The primary causes of past reportable spills at Fort Cavazos were human error and equipment failure.

3.4.1 CATASTROPHIC EVENTS

Although unlikely, a catastrophic event such as a tornado, violent storm, explosion, or catastrophic failure of a tank at a large facility may result in the discharge of large quantities of oil or fuel.

3.4.2 RISK REDUCTION FACTORS

The following practices and factors reduce the risk of discharges occurring at Fort Cavazos's oil and fuel storage facilities:

a. Routine operational inspections and periodic scheduled formal in-service and out of service inspections and testing of the DFSP oil storage and transfer equipment,

b. Daily visual inspections by facility personnel,

c. Routine maintenance by DPW personnel,

d. Standard operating procedures for transfer and re-circulating activities,

e. Preventive measures such as valve lockouts at some facilities,

f. Visible placement of warning signs (e.g. "Flammable," "No Smoking w/in 50 ft"),

g. Sufficient secondary containment for most facilities (reducing the impact of a discharge on the surrounding environment).

h. Prompt removal of water from secondary containment structures,

i. ASTs and USTs are generally less than fifteen years old,

j. Replacement of existing shop-fabricated tanks at oil/fuel storage and collection

3.5 FACILITY REPORTABLE SPILL HISTORY

Fort Cavazos has averaged three (3) reportable spills per year over the last five (5) years with none exceeding 500 gallons. All spills were contained at the initial spill site, cleaned up, and the site was returned pre-spill conditions. A comprehensive electronic spill database is maintained by the DPW Environmental Division to track reportable and non-reportable spill response activities. Spill records can be provided to regulatory agencies upon request.

4. DISCHARGE SCENARIOS

F-24 aircraft fuel, or gasoline would be the most likely products accidentally discharged at this Installation. The spill potential for other petroleum products is much less, based on the relative quantities maintained at the Installation. Volumes for small, medium, and worst-case discharges, as defined in the EPA final rules, have been identified or calculated for the Installation's significant aboveground bulk fuel facilities. Required distance calculation worksheets in Section 8 of this plan of this plan. Fort Cavazos has identified the response resources necessary to respond to small, medium, and worst-case spills. A table of available resources is in Section 1.4 of the ERAP. Fort Cavazos also ensures the availability of the additional response resources for these spills, by contract or other approved means as described in 40 CFR 112. Information for contract organization is available in Table 1-4 Emergency Response Contractors in Section 1.5.3 of the ERAP. All response equipment is designed to function in the operating environment at Fort Cavazos.

4.1 SMALL DISCHARGES

A small discharge is defined as any discharge of 2,100 gallons (50 barrels) or less, provided that this amount is less than the worst-case discharge amount as defined by EPA.

4.1.1 SMALL DISHARGE PROBABLE LOCATION

A small discharge would most likely occur in or near POL storage areas, fuel transfer areas, on roadways, or in the training areas during training exercises.

4.1.2 SMALL DISCHARGE PROBABLE CAUSE

A small spill is typically caused by vehicle accidents, equipment failure or leaks, or operator error while managing POL storage areas or fuel transfer operations.

4.1.3 SMALL DISCHARGE CONTAINMENT

Small discharges would occur in the immediate area of the failure. A spill from an AST or drum would be contained within the secondary containment system. A spill from an accident or fuel transfer operation would be contained by trained personnel with the response equipment listed in section 1.3.

4.1.4 POTENTIAL FOR ENVIRONMENTAL IMPACT FROM SMALL DISCHARGES

A small discharge in contained areas should not cause significant environmental impact. The containment will hold the spill until free product can be recovered and/or cleanup operations can begin (cleanup operations typically entail removal and replacement of contaminated soil or stone, or cleanup of concrete containment surfaces).

4.2 MEDIUM DISCHARGES

A medium discharge is defined as any discharge up to 36,000 gallons or 10 percent of the worst-case discharge, whichever is less (40 CFR Part 112, Appendix F). Since 36,000 gallons is less than 10 percent of the worst-case discharge, the medium discharge for the base will be defined as any discharge between 2,100 and 36,000 gallons.

4.2.1 PROBABLE LOCATION

The most probable location for a spill of this size to occur would be at one of the three DFSPs or one of the Fort Cavazos, end-use established oil transfer stations. The most likely scenario for a spill of this magnitude would occur during refilling of one of the major aboveground storage tank systems. Refilling occurs through tanker trailer-type delivery vehicles.

4.2.2 PROBABLE CAUSE

e. The RGAAF, YDAHP RRF, the Main BFSF, or P2 Services: Failure of tankers and loading/unloading operations may result in a maximum spill of 5,000 gallons. Rupture of tanks or pipes may cause reportable spills. Failure of pumps, pipes, transfer lines, or other transfer equipment also may cause a reportable spill.

f. Post Exchange Service Centers (AAFES): Rupture of tanks or pipes may cause reportable spills. Failure of pumps, pipes, transfer lines, or other transfer equipment also may cause a reportable spill. Loading/unloading operations may result in a maximum spill of 5,000 gallons.

g. NFC FARP: Rupture of tanks or pipes may cause reportable spills. Failure of pumps, pipes, transfer lines, or other transfer equipment also may cause a reportable spill. Loading/unloading operations may result in a maximum spill of 5,000 gallons.

4.2.3 CONTAINMENT

All major tank systems have secondary containment, so a leak from the tank itself would not be expected to create a release of this magnitude. Refueling areas are constantly monitored by transfer operations personnel, and the spill would not likely be of any significant volume. Facility piping is visually monitored during transfers into the AST systems. Delivery piping to equipment is predominantly located underground and would not be expected to cause a surface spill.

4.2.4 POTENTIAL FOR IMPACT TO THE ENVIRONEMENT

A spill in this category at any facility or occurring in the training areas could travel some distance before full containment can be achieved. The spill would most-likely impact vegetation, fish, and wildlife in and around the affected areas, natural drainage features, and streams. The spill pathway from some of these areas may pass near residential areas, school, or childcare facilities that could be affected by fumes.

4.3 WORST-CASE DISCHARGE SCENARIO

The most probable worst-case discharge is a catastrophic failure of the largest bulk storage tank (Tank 111) with a capacity of 636,488 gallons. This tank is located at the BFSF.

4.3.1 PROBABLE LOCATION

The most probable location for a spill of this size to occur would be at Main Cantonment Bulk Fuel Storage Facility.

4.3.2 PROBABLE CAUSE

Major risk scenarios are based on the inherent risk to a military establishment: an attack on the base during wartime conditions, a terrorist attack on the base, a natural disaster, or an aircraft crash in the bulk oil storage area. Two natural events could cause a major impact on the oil storage and distribution system: lightning strikes causing an explosion and fire or severe weather. Due to U. S. Army and Fort Cavazos operational inspection requirements for equipment and POL storage containers, no inherent risk of discharge due to age or a chain reaction at Fort Cavazos exists.

4.3.3 CONTAINMENT

All of the major tank systems have secondary containment, so a leak from the tank itself would not be expected to create a release of this magnitude. In the event of an incident of this magnitude, containment efforts would be secondary to protection of life health and safety. Containment efforts would be initiated after any threat (severe weather, fire/explosion, terrorist activity, etc…) has been eliminated which could cause a delay in and increase challenges in containing the spill. A spill of this magnitude would require the use of an OSRO.

4.3.4 POTENTIAL FOR IMPACT TO THE ENVIRONEMENT

While extremely unlikely, a spill in this category could travel some distance before full containment can be achieved. The spill would most-likely impact vegetation, fish, and wildlife in and around the affected areas, natural drainage features, streams, and lakes and could potentially the reach Belton Lake and impact the local drinking water supply. The spill pathway from some of these areas may pass near residential areas, school, or childcare facilities that could be affected by fumes.

5. FACILITY RESPONSE PLAN IMPLENTATION

5.1 PHASE I – DISCOVERY, EVALUATION, AND INITIAL NOTIFICATION

The following section describes the implementation process to be followed in the event of a spill during loading and unloading operations, bulk fuel transfer operations, AST, and storage container management, or during vehicle/equipment operations.

5.1.1 DISCHARGE DISCOVERY

d. Monitoring/Inspection - All tank truck loading and unloading operations, including bulk fuel transfer operations at the DFSP facilities are manned and supervised operations. ASTs and other storage containers at Fort Cavazos are visually inspected by personnel once per day to identify any leaks or spills and any problems or deficiencies with emergency response/safety equipment. A monthly inspection is conducted in accordance with inspection procedures identified in the Fort Cavazos SPCCP. The facility manager will be informed of any deficiencies noted during the inspection. In the event of discovering a spill or leak, spill response activities will be initiated.

e. Automated Discharge Detection - All DFSP bulk fuel storage ASTs have automatic tank gauge systems to detect discharges or leaks. These ASTs are all within sufficiently impervious secondary containment dikes to facilitate detection of any aboveground discharges or leaks from the tanks and appurtenances in the dikes.

The RGAAF South Ramp hydrant's buried stainless-steel supply and return pipelines and lateral lines have a leak detection with alarm system. The hydrant piping, including the hydrant, surge suppression, and isolation valve pits, is contained in a trench lined with a fuel-resistant 20-mil Hypalon liner. A perforated Polyvinyl Chloride (PVC) line along the trench contains a leak-detection cable for any fuel collected in the lined trench. Fort Cavazos service stations have automatic spill detection systems which sound an alarm to alert the attendant when a spill occurs at an AST or UST. The ASTs and USTs are double-walled tanks with interstitial monitors that sound an alarm when detecting fuel in the interstitial space. Following activation of an alarm, the attendant, or other qualified inspector will immediately conduct a visual inspection of the tank and sample the interstitial space using a sample tube.

5.1.2 EVALUATING THE SPILL SITE FOR ORGANIZATIONAL REPORTING REQUIREMENTS

Personnel discovering a discharge/spill incident must evaluate the site to determine if a potential threat to public health or safety exists prior to taking and response action. Potential risks could be but are not limited to: An unknown substance; fire, electrical, or explosion hazard; mix of incompatible material or toxic substances; confined space; structural damage/building collapse; vehicle or equipment accident site that has not been secure. Do not approach a discharge/spill incident site if a potential for public health or safety risk exists. Only give assistance or first aid to injured persons if it is safe to do so and call the FES 911 immediately for medical assistance. If it is determined that no immediate threat exists, evaluate the discharge/spill incident according to the following reporting criteria:

a. Is the release more than five (5) gallons of petroleum products?

b. Is it an unknown substance?

c. Has the spilled reached a surface water drainage system (stormwater conveyance, stream, creek, river, pond, lake, etc…) or cause a visible sheen on surface water?

d. Is the spill located in a confined space?

5.1.3 INITIAL REPORTING

If the spill does not meet any of the above reporting criteria, the site supervisor/commanding officer will be notified, and unit/organization will initiate containment and clean-up activities and prescribed in Sections 5.2 and 5.3 below. If the spill is discovered in a cantonment area and meets one or more of the above reporting criteria, immediately notify the spill site supervisor/commanding officer, and FES, Fire Dispatch by calling 911. For spills occurring in the Fort Cavazos Training Areas, contact the DPTMS Range Operations office at (254) 287-3130. Once on site the FES will evaluate the spill and notify the DPW Environmental office if further assistance is needed and to ensure that any local, state, or federal agency notifications are made within regulatory reporting parameters.

5.2 PHASE II CONTAINMENT AND COUNTERMEASURES

5.2.1 ORGANIZATION RESPONSE FOR REPORTABLE SPILLS

If the spill meets the criteria of a reportable spill as outlined in 5.1.2, the discovering unit/organization will call 911, to report the spill. The IRT will evaluate the spill and initiate containment and countermeasures as outlined in the ERAP, Section 1.8 Immediate Action for Emergency Spill.

5.2.2 ORGANIZATION RESPONSE FOR NON-REPORTABLE SPILLS

The following action will be initiated by the site supervisor/commanding officer of the spilling organization if the release does not meet any of the criteria in 5.1.2, then take these steps:

a. Identify and Stop the Spill at the Source – Act quickly to secure pumps, close valves, close spill drains, tighten gaskets, etc.

b. Isolate the Spill – Enforce safety and security measures and secure the area. Activate emergency alarm systems if necessary. Keep non-essential persons away until the spill has been cleaned-up.

c. Shut off ignition Sources – Ensure any source of ignition such as motors, electrical circuits, open flames, etc… are removed or extinguished.

d. Initiate Containment – place absorbent booms or sweepable absorbents provided in unit/organization spill kits, along the leading edges of the spill to prevent further spreading and protect storm drains and other stormwater conveyances.

5.2.3 CONTAINMENT AND RECOVERY CAPABILITIES

The DPW Environmental Response Team will coordinate any additional installation support such as: Earth moving equipment and personnel to clear recovery paths and erect earthen berms; vehicles and equipment needed to recover free board contained freeboard product; or available storage tanks for temporary storage of recovered product until disposal can be arranged. Available resources are identified in Table 1-2 Spill Equipment List by Location of this plan. If the resources required to contain the spill exceeds Fort Cavazos capabilities, a request for OSRO support will be initiated by the QI. If OSRO support is required, the OSRO will complete Phases II thru IV of this Section.

5.3 PHASE III CLEAN-UP AND DISPOSAL

Once the IRT has mitigated any threats to human health and safety and the spill has been contained, it is the responsibility of the unit/organization to clean-up the spill. DPW Environmental Response Team personnel will guide and assist the unit and ensure that necessary resources are available.

5.3.1 SPILL CLEAN-UP

Spill Clean-up materials and equipment for incidental not-reportable spills are maintained in spill kits at all unit/organization maintenance facilities for use in spill response. If the resources required to clean a spill exceeds the unit/organization capabilities, the DPW Environmental Response Team will provide additional materials and/or equipment as needed or assist in requesting support from other internal Fort Cavazos agencies as needed. The unit/organization will:

a. Apply thin layer of sweepable absorbents on hard surfaces or oil absorbent pads for pooled liquids or free-board product on waterbodies. Allow absorbents to stay in place for sufficient time to allow for complete absorption.

b. Once liquids have been absorbed, collect and containerize all contaminated materials. When sweeping up absorbent materials, use brooms and/or non-sparking shovels to collect contaminated materials to minimize the potential for spark/fire.

c. Separate contaminated sweepable absorbents from absorbent pad/booms and store in plastic bags provided in spill kits or by DPW Environmental Personnel. Bags of contaminated materials should be kept in the Non-Hazardous Waste Accumulation Point (NHWAP) prior to disposal.

5.3.2 EXCAVATION OF CONTAMINATED SOIL

If an OSRO has not been initiated for spill clean-up, It is the responsibility of the spilling unit/organization to excavate any soils that have been contaminated as a result of the spill. If the spilling organization does not have the resource to excavate soils, DPW Environmental Response Personnel will request support from DPW O&M Roads and Grounds. The following steps are required for soil excavation.

a. Prior to soil excavation, the unit/organization must initiate an emergency line locate. To initiate an emergency line-locate, Contact the DPW Operations and Maintenance at (254) 535-2993, Request an Emergency Line Locate for Spill Response and provide the spill site location and a call back number.

b. Once all lines have been identified, the unit/organization can proceed with excavation of contaminated soils.

c. Depending on the amount of soil excavated, the contaminated soil can be placed in a plastic bag, a metal, open head, drum, or a covered trailer for disposal. Plastic bags and drums should be kept in the NHWAP prior to turn-in.

5.3.3 DISPOSAL OF CONTAMINATED SOIL AND MATERIALS

Coordinate through the classification unit to dispose of all contaminated soils and materials. The unit/organization will need to make and appointed for turn-in and submit the completed the 3161 provided by the CU Yard. On the day of appointment, the unit/organization should have two (2) printed copies of the 3161 on hand.

5.4 PHASE IV SITE RESTORATION

Once the clean-up phase is complete, the Unit/organization is responsible for restoring the spill site, to the original condition. If backfill soils are required for restoration, the unit will call DPW O&M roads and grounds to request a pick-up location or delivery of backfill soils. Part of the restoration process includes re-seeding or providing vegetation stabilization to the area. Once the area has been restored and seeded, call DPW Environmental Response Team to provide a closeout inspection of the impacted area.

5.5 REVIEW AND UPDATE OF THE FACILITY RESPONSE PLAN

Fort Cavazos will review the Facility Response Plan annually (at a minimum) in conjunction with the annual review of the Fort Cavazos SPCC Plan. The plan will be updated under the following conditions: If plan improvements are identified as a result of an post-exercise debriefing; if potential issues or improvements are identified as a result of an emergency spill situation; as a result of organizational, procedural, manpower, or equipment changes; as a result of changes identified during the annual review.

6. DISPOSAL OF RECOVERED MATERIAL

This section describes how and where Fort Cavazos intends to recover, reuse, decontaminate, or dispose of materials after a discharge has occurred.

6.1 FORT CAVAZOS STORAGE OR DISPOSAL FACILITITES

Fort Cavazos has three storage facilities that can provide temporary storage for recovered or contaminated materials and one disposal facility for final disposition of nonhazardous materials. The Fort Cavazos Facilities are:

6.1.1 FORT CAVAZOS POLLUTION PREVENTION (P2) SERVICES

The Fort Cavazos P2 Services Facility is at building 1949 (37th St. and Murphy Drive). This Program is responsible service of all PSTs and OWSs on the installation and collection, separation, and storage of all used or off-spec POL products, as well as POL contaminated wastewater, and providing temporary storage of product for recycle or disposal. This facility can provide vehicles and trained personnel for extraction of freeboard product (amount of support is dependent on availability of equipment and qualified operators at the time of the spill). The facility can also provide at least 10,000 gallons of storage for recovered product (potential for additional storage at this location is dependent on current operations).

6.1.2 FORT CAVAZOS CLASSIFICATION UNIT (CU)

The Fort Cavazos CU is registered with the TCEQ as a Central Accumulation Area. It is located at Building 1348, 37th Street and Ivy Division Rd. The CU is a multipurpose operation for managing HAZMAT and waste, reusable material, recyclable materials, and other associated products. Hazardous waste generated by Fort Cavazos units, tenants, and contractors are accepted, classified, repackaged, labeled, stored, and processed for disposal.

6.1.3 BIOREMEDIATION FACILITY

The bioremediation facility is located within the P2 Services fenced area. All contaminated bio-degradable sweepable absorbent and excavated soils will be transferred to this location for processing.

6.1.4 FORT CAVAZOS MUNICIPAL LANDFILL

The Fort Cavazos Municipal Landfill is a government owned/contract operated facility located on Turkey Run Road. All non-hazardous materials recovered from a spill site, will be sent here for final disposition. This includes non-hazardous absorbent pads/booms, PPE, equipment, spent chemicals, and used decontamination solutions. Waste manifests are not required.

6.2 OFF-INSTALLATION DISPOSAL PROCEDURES

There are two primary mechanisms in place for disposal of recovered materials that must be transported off Fort Cavazos for final disposition.

6.2.1 QUALIFIED RECYCLE PROGRAM (QRP)

The Fort Cavazos QRP has a contract in place for transporting all used oil to a qualified recycling facility for final disposition. Waste manifests are not required.

6.2.2 DEFENSE LOGISTICS AGENCY (DLA) DISPOSITION SERVICES

The DLA Disposition Services has a contract in place for both qualified recycling facilities (for off-spec fuel and used antifreeze) and hazardous waste storage and disposal facilities for final disposition of hazardous waste. This includes hazardous absorbent pads/booms, PPE, equipment, spent chemicals, and used decontamination solutions. Wastes are manifested at the CU Facility prior to transfer.

7. CONTAINMENT AND DRAINAGE PLANNING

This section describes the Fort Cavazos plan for containment and controlling a spill through drainage. It primarily describes containment and drainage of POL in the event of a release.

7.1 CONTAINMENT VOLUME

All ASTs on Fort Cavazos have sufficient secondary containment for tank volume. Precipitation that accumulates in secondary containment systems will not be released until it has been examined to ensure no oil contamination will be discharged with the water. Fort Cavazos operators and users, including DFSP contractor, personnel perform daily inspection of containment equipment. In addition to regular maintenance, these personnel provide expedient repair to any leaks or other problems potentially causing contamination of contained water.

7.1.1 STORAGE TANKS AT THE BFSF

The six bulk storage tanks in the BFSF are within concrete dike containment areas. Each containment area has a sump and drainage piping for removing precipitation from within the dike. Outlet valves on this piping are kept closed and locked. After a storm event, water within the containment area is checked for surface sheen or other evidence of oil before the outlet valves are opened to drain uncontaminated storm water. If no evidence of oil is present, this water is discharged to the storm water drainage that eventually flows into Clear Creek, House Creek, and Cowhouse Creek.

7.1.2 LOADING AND UNLOADING AREAS

Secondary containment is provided for the tank truck loading/unloading racks and stands at the BFSF. The tank truck load racks and tank truck unload stands have concrete paved and curbed containment pads constructed with gravity quick drainage Systems. Below ground drain pipes connect the rack and stand pad sump drains to an OWS or to the OWS recovery/spill tank (AST OWS) in the event of an oil spill at a rack or stand. The capacity of AST OWS, 10,000 gallons, is sufficient to hold the entire content of the largest tank truck compartment loaded at the BFSF racks, which is 8,000 gallons. Outflow from the OWS is discharged into the storm water drainage system that eventually flows into Clear Creek, House Creek, and Cowhouse Creek

7.2 STORMWATER DRAINAGE

Section 1.9 Facility Diagrams of the ERAP provides a full description and diagrams of Fort Cavazos stormwater drainage routes.

7.2.1 DRAINAGE TROUGHS CONSTRUCTION MATERIAL

The construction materials used in drainage troughs are concrete, asphalt, earthen, and typical catch basin and piping infrastructure.

7.2.2 DRAINAGE SYSTEM SEPARATORS

Secondary containment at bulk fuel facilities is drained with manually operated, gravity feed, discharge pipes. These valves are normally maintained in a locked and closed position. Storm water is drained from secondary containment as necessary in accordance with the guidelines found in the SPCC plan.

7.2.3 BOOM/WEIR CONTAINMENT CAPACITY

Tables 7-1 Bridges and Overpasses, identify bridges and overpasses crossing the potentially affected waterways for the worst-case discharges. Their possible uses are: access, “lookout” points, and potential creek booming locations. These points are generally accessible by motor vehicle for deployment of booms, skimmers, personnel, and other necessary response equipment.

Table 7-1 Bridges and Overpasses

Bridge/ Overpass	Accessibility	Waterway	Distance in Miles from BFSF
Oakalla Road	Access point is on northwest side of Oakalla Road and southwest of Reese Creek. Four-wheel drive access to creek edge or walk 100 feet north of road to reach creek. Low flow area.	Reese Creek	~ 2 (RGAAF)
Maxdale Road	Access point is on southeast side of Maxdale Road and northeast of Reese Creek. Truck may park at this point and access to creek is a short distance. Low flow area. Creek may be forded next to bridge.	Reese Creek	~ 3 (RGAAF)
Pump Station Road	Access points on either side of Pump Station Road.	Clear Creek	~ 1.5 (RGAAF)
I-14	Access points on either side of I-14.	Clear Creek	~ 3.7 (RGAAF)
Tank Destroyer Road	Access point on south side of Tank Destroyer Road.	Clear Creek	~ 1.8 (BFSF) ~ 4.3 (RGAAF)
Copperas Cove Road	Access point on east of Clear Creek. Parking available near barricade. Clear Creek is a pool approximately 20 feet wide at this location.	Clear Creek	~ 2.2 (BFSF) ~ 4.7 (RGAAF)
Turkey Run Road	Access point on north side of Turkey Run Road on both the east and west sides of Clear Creek. Clear Creek is approximately 8 feet wide at this point and flowing.	Clear Creek	~ 3.9 (BFSF) ~ 6.4 (RGAAF)
West Range Road	Access point on east side of W. Range Road approximately 300 feet south of House Creek. House Creek is approximately 40 feet wide and flowing at this point.	House Creek	~ 9.4 (BFSF)
Warrior Way	Access point at YDAHP RRF inside of fence on Fort Cavazos property.	Drainage Basin	~ 0.2 (YDAHP)
Duncan Avenue	Access point at Duncan Avenue or N 24th street into the concrete channel.	Concrete-lined Channel	~ 0.5 (YDAHP)
East Rancier Avenue (493)	Access point on south side of Route 439.	Unnamed Tributary to South of Nolan Creek	~ 0.8 (YDAHP)
Harris Avenue	Access point on either side of the bridge. (Private Property)	Unnamed Tributary to	~ 0.9 (YDAHP)

		South of Nolan Creek	
Parmer Avenue	Access point on either side of the bridge. (Private Property)	Unnamed Tributary to South of Nolan Creek	~ 1 (YDAHP)
Greenwood Avenue	Access point on either side of the bridge. (Private Property)	Unnamed Tributary to South of Nolan Creek	~ 1.3 (YDAHP)
East Avenue G	Access point on either side of the bridge. (Private Property)	Unnamed Tributary to South of Nolan Creek	~ 1.4 (YDAHP)
South 28th Street	Access point on either side of the bridge. (Private Property)	Unnamed Tributary to South of Nolan Creek	~ 1.5 (YDAHP)
South W.S Young Drive	Access point on south side of W.S Young Avenue	South Nolan Creek	~ 1.8 (YDAHP)
South 38th Street	Access point on SW or NW side of 38th Street	South Nolan Creek	~ 2.6 (YDAHP)
South Twin Creek Drive	Access point on both sides of grassy terrain.	South Nolan Creek	~ 3.6 (YDAHP)
North Roy Reynolds Drive	Access point on both sides of grassy terrain.	South Nolan Creek	~ 4.7 (YDAHP)
North Amy Lane	Access point on both sides of the bridge.	South Nolan Creek	~ 5.7 (YDAHP)
Buffalo Trail	Access point NE side via dirt road.	South Nolan Creek	~ 6.5 (YDAHP)
I-14	Access point on both sides of the bridge	South Nolan Creek	~ 8.2 (YDAHP)
Old Nolanville Road	Access point on SW side via dirt road.	South Nolan Creek	~ 9.4 (YDAHP)

8. PLANNING DISTANCE CALCULATIONS

Per Section 5 of Attachment C-III to Appendix C of 40 CFR 112, once it is determined that the facility could cause substantial harm, the owner must evaluate the potential for portions of a worst-case discharge transported over land to navigable waters of the United States. In accordance with paragraph 1.6 of Attachment C-III of 40 CFR 112 this evaluation will reference the response times discussed in Appendix E of 40 CFR 112. The response time for a Tier 1 response is 12 hours. Fort Cavazos worst case discharge is described in Section 4, Discharge Scenarios, of this plan. Fort Cavazos spill pathways are outlined in Section 1.9 Facility Diagrams of the ERAP. Evaluations were conducted for the three DFSPs (Main Cantonment BFSF, YDAHP RRF, RGAAF), as these are the most probable locations for a spill of this size.

8.1 PLANNING DISTANCE WORKSHEET

The planning distance worst-case discharge worksheet and formulas used for evaluations are based on the processes, requirements, and formula found in Section 2 of Attachment C-III to Appendix C to 40 CFR 112 for oil transport on moving navigable waters.

Part I. Background information: largest tank with adequate containment (Tank 111 in the BFSF).

Step (A). Calculate worst-case discharge in gallons (in accordance with Appendix D to 40 CFR 112) $\boxed{636{,}888}$

Step (B). Oil group[1]{1} (from Table 3 to Appendix E of 40 CFR 112) $\boxed{1}$

Step (C). Operating area (choose one) $\boxed{x}$ Nearshore/Inland or Rivers

$\boxed{}$ Great Lakes and Canals

Step (D). Percentages of oil (from Table 2 to Appendix E of 40 CFR 112)

Percentage Lost to Natural Dissipation	Percentage Recovered Floating Oil	Percent Oil Onshore
80	10	10
(D1)	(D2)	(D3)

Step (E1). On-water oil recovery: $\dfrac{\text{Step (D2) x Step (A)}}{100}$ $\boxed{63.688}$ (E1)

Step (E2). Shoreline recovery: $\dfrac{\text{Step (D3) x Step (A)}}{100}$ $\boxed{63{,}688}$ (E2)

[1] *Group I oils represents > 97.5% of total oil storage capacity at Fort Cavazos. Therefore, only Group 1 spills have been considered.*

Step (F). Emulsification factor
(From Table 3 to Appendix E of 40 CFR 112)

1.0
(F)

Step (G). On-water oil recovery resource mobilization factor
(From Table 4 to Appendix E of 40 CFR 112)

Tier 1	Tier 2	Tier 3
0.30	0.40	0.60
(G1)	(G2)	(G3)

Part II. On-water oil recovery capacity (gallons/day)

Tier 1	Tier 2	Tier 3
19,107	25,475	38,213
Step (E1) x Step (F) x Step (G1)	Step (E1) x Step (F) x Step (G2)	Step (E1) x Step (F) x Step (G3)

Part III. Shoreline cleanup volume (gallons)

63.688
Step (E2) x Step (F)

Part IV. On-water response capacity by operating area
(From Table 5 to Appendix E of 40 CFR 112)
(Amount needed to be contracted for in gallons/day)

Tier 1	Tier 2	Tier 3
78,750 gals/day	157,500 gals/day	315,000 gals/day
(J1)	(J2)	(J3)

Part V. On-water amount needed to be identified, but not contracted for in advance (barrels/day)

Tier 1	Tier 2	Tier 3
0	0	0
Part II Tier 1 Step (J1)	Part II Tier 2 Step (J2)	Part II Tier 3 Step (J3)

8.2 PLANNING DISTANCE FORMULAS

The planning distance worst-case discharge worksheet and formulas used for evaluations are based on the processes, requirements, and formula found in Section 2 of Attachment C-III to Appendix C to 40 CFR 112 for oil transport on moving navigable waters.

8.2.1 VELOCITY FORMULA

The formula to calculate the planning distance for oil transport on moving navigable waters is:

D = V x T x C, where

D = the distance downstream from a facility within which fish and wildlife and sensitive environments could be injured.

V = the velocity of the river (in ft/sec) as determined by Chezy-Manning's equation.

T = the time interval specified in corresponding time tables.

C = constant conversion factor: 0.68 sec-mile/hr-ft.

8.2.2 CHEZY - MANNING EQUATION

The Chezy-Manning Equation is used to determine velocity. The computed velocity would approximate the velocity of the watercourse for high flows at or near flood stage. The formula is:

V = $\mathbf{C\ R^{1/2} \times S^{1/2}}$**,** where

N = Manning's Roughness Coefficient from Table 8-1 (Section 8.2.3 of this document).

R = the hydraulic radius, which can be approximated for parabolic channels by multiplying the average mid-channel depth of the river (in feet) by 0.667.

S = the average slope of the river (unitless), which was calculated from the elevation features on Google Earth by subtracting the closest downstream elevation from the closest discharge elevation and dividing this by the distance between each point.

8.2.3 MANNING'S ROUGHNESS COEFFICIENT

Table 8-1 below provides the formula for Manning's Roughness Coefficient for Natural Streams.

Table 8-1 Manning's Roughness Coefficient for Natural Streams	
Stream Description	**Roughness Coefficient (N)**
Minor streams (Top width < 100 ft.)	
Clean:	
Straight	0.03
Winding	0.04
Sluggish (weedy, deep pools)	
No trees or brush	0.06
Trees and/or brush	0.10
Major Streams (Top width > 100 ft.)	
Regular section:	
(No boulders/brush)	0.035
Irregular section:	
(Boulders or brush)	0.05

8.3 OIL TRANSPORT OVER LAND

Evaluations were conducted for the three DFSPs (Main Cantonment BFSF, YDAHP RRF, RGAF), as these are the most probable locations for a spill of this size.

8.3.1 CALCULATION TABLES FOR THE MAIN CANTONMENT BFSF

This section provides the calculation used to obtain planning distance for the Main Cantonment BFSF. The calculations include Discharge Information, Hydraulic Radius, and discharge velocity.

a. Table 8-2 Main Cantonment BFSF Discharge Information

Table 8-2 Main Cantonment BFSF Discharge Information					
Route	**Distance (Miles)**	**Starting Elevation**	**Ending Elevation**	**Slope (S) [1]**	**Roughness Coefficient (N) [2]**
From AST 111 to Clear Creek via a seasonally dry unnamed tributary	1.7	960	840	0.0134	0.10
From Clear Creek to House Creek.	4.7	840	736	0.0042	0.05
From House Creek to Cowhouse Creek	4.6	736	656	0.0033	0.05
From Cowhouse Creek to Belton Lake	13.2	656	594	0.0009	0.05
Total Distance	**24.2**				
Notes: [1]Slope = (Starting Elevation – Ending Elevation)/Distance (in feet). [2]Roughness Coefficient = estimated average from Table IRP 11-1 for the route.					

b. Table 8-3 BFSF Hydraulic Radius

Table 8-3 Hydraulic Radius		
Route	Estimated average mid-channel depth	Hydraulic radius (R) [1]
From AST 111 to Clear Creek via a seasonally dry unnamed tributary	1.5	1.00
From Clear Creek to House Creek.	2.0	1.33
From House Creek to Cowhouse Creek	2.5	1.67
From Cowhouse Creek to Belton Lake	3.0	2.00
Note: [1]Hydraulic radius is approximated by multiplying mid-channel depth by 0.667.		

c. Table 8-4 BFSF Discharge Velocity

Table 8-4 BFSF Discharge Velocity 1.429 feet/second X 4.8 hours X 0.68 (conversion factor) = **4.7 miles**				
Route	N	R	S	V [1]
From AST 111 to Clear Creek via a seasonally dry unnamed tributary	0.10	1.00	0.0134	1.736
From Clear Creek to House Creek.	0.05	1.33	0.0042	2.352
From House Creek to Cowhouse Creek	0.05	1.67	0.0033	2.426
From Cowhouse Creek to Belton Lake	0.05	2.00	0.0009	1.429
Note: [1]Velocity (V) = 1.5/N x $R^{2/3}$ x $S^{1/2}$ in feet/second.				

8.3.2 CALCULATION TABLES FOR THE RGAF

This section provides the calculation used to obtain planning distance for the GRAF. The calculations include Discharge Information, Hydraulic Radius, and discharge velocity.

a. Table 8-5 RGAF Discharge Information

Table 8-5 RGAF Discharge Information					
Route	Distance (Miles)	Starting Elevation	Ending Elevation	Slope (S) [1]	Roughness Coefficient (N) [2]
From AST 2 to Reese Creek via a seasonally dry unnamed tributary	0.05	963	956	0.0255	0.10
From unnamed tributary to Reese Creek.	2.0	956	876	0.0076	0.05
From Reese Creek to Lampasas River	7.2	876	685	0.0050	0.05
From Lampasas River to Stillhouse Hollow Lake	18.7	685	622	0.0006	0.04
Total Distance	**27.95**				

Notes:
[1]Slope = (Starting Elevation – Ending Elevation)/Distance (in feet).
[2]Roughness Coefficient = estimated average from Table IRP 11-1 for the route.

b. Table 8-6 RGAF Hydraulic Radius

Table 8-6 RGAF Hydraulic Radius		
Route	**Estimated average mid-channel depth**	**Hydraulic radius (R) [1]**
From AST 2 to Reese Creek via a seasonally dry unnamed tributary	1.0	1.00
From unnamed tributary to Reese Creek.	2.0	1.33
From Reese Creek to Lampasas River	2.5	1.67
From Lampasas River to Stillhouse Hollow Lake	4.0	2.67
Note: [1]Hydraulic radius is approximated by multiplying mid-channel depth by 0.667.		

c. Table 8-7 RGAF Discharge Velocity

Table 8-7 RGAF Discharge Velocity 1.768 feet/second X 7.6 hours X 0.68 (conversion factor) = **9.1 miles**				
Route	**N**	**R**	**S**	**V [1]**
From AST 2 to Reese Creek via a seasonally dry unnamed tributary	0.10	1.00	0.0255	2.395
From unnamed tributary to Reese Creek.	0.05	1.33	0.0076	3.163
From Reese Creek to Lampasas River	0.05	1.67	0.0050	2.986
From Lampasas River to Stillhouse Hollow Lake	0.04	2.67	0.0006	1.768
Note: [1]Velocity (V) = 1.5/N x $R^{2/3}$ x $S^{1/2}$ in feet/second.				

8.3.3 CALCULATION TABLES FOR THE YDAHP

This section provides the calculation used to obtain planning distance for the YDAHP RRF. The calculations include Discharge Information, Hydraulic Radius, and discharge velocity.

a. Table 8-8 YDAHP RRF Discharge Information

Table 8-8 YDAHP RRF Discharge Information					
Route	Distance (Miles)	Starting Elevation	Ending Elevation	Slope (S) [1]	Roughness Coefficient (N) [2]
From AST 1 to seasonally dry drainage basin	0.2	870	860	0.0095	0.10
From seasonally dry drainage basin through concrete-lined channel to seasonally dry unnamed tributary	0.6	860	837	N/A [3]	N/A [3]
Seasonally dry unnamed tributary to South Nolan Creek	0.8	837	799	0.0090	0.05
From South Nolan Creek to Nolan Creek	15.6	799	573	0.0027	0.05
From Nolan Creek to Leon River	9.5	573	442	0.0026	0.05
From the Leon River to Lampasas River	8.1	442	425	0.0004	0.04
Total Distance	**26.6**				

Notes:
[1]Slope = (Starting Elevation – Ending Elevation)/Distance (in feet).
[2]Roughness Coefficient = estimated average from Table IRP 11-1 for the route.
[3]As per section 5.2 of Appendix C to 40 CFR 112, flow through concrete channels can range 3–25 feet per second. An average of 14 feet per second will be used.

b. Table 8-9 YDAHP RRF Hydraulic Radius

Table 8-9 YDAHP RRF Hydraulic Radius		
Route	Estimated average mid-channel depth	Hydraulic radius (R) [1]
From AST 1 to seasonally dry drainage basin	1.5	1.00
From seasonally dry drainage basin through concrete-lined channel to seasonally dry unnamed tributary	N/A [2]	N/A [2]
Seasonally dry unnamed tributary to South Nolan Creek	2.0	1.33
From South Nolan Creek to Nolan Creek	2.5	1.67
From Nolan Creek to Leon River	2.5	1.67
From the Leon River to Lampasas River	4.0	2.67

Notes:
[1]Hydraulic radius is approximated by multiplying mid-channel depth by 0.667.
[2]As per section 5.2 of Appendix C to 40 CFR 112, flow through concrete channels can range 3–25 feet per second. An average of 14 feet per second will be used.

c. Table 8-10 YDAHP RRF Discharge Velocity

Table 8-10 YDAHP RRF Discharge Velocity 2.154 feet/second X 1.0 hours X 0.68 (conversion factor) = **1.46 miles**				
Route	**N**	**R**	**S**	**V [1]**
From AST 1 to seasonally dry drainage basin	0.10	1.00	0.0095	1.462
From seasonally dry drainage basin through concrete-lined channel to seasonally-dry unnamed tributary [2]	N/A	N/A	N/A	14.0
Seasonally dry unnamed tributary to South Nolan Creek	0.05	1.33	0.0090	3.442
From South Nolan Creek to Nolan Creek	0.05	1.67	0.0027	2.195
From Nolan Creek to Leon River	0.05	1.67	0.0026	2.154
From the Leon River to Lampasas River	0.04	2.67	0.0004	1.444
Note: **1**Velocity (V) = 1.5/N x $R^{2/3}$ x $S^{1/2}$ in feet/second **2**As per section 5.2 of Appendix C to 40 CFR 112, flow through concrete channels can range 3–25 feet per second. An average of 14 feet per second will be used.				

8.4 TOTAL PLANNING DISTANCE

The Evaluation results and total planning distance for all three locations are are identified in Table 8-11 Total Planning Distance.

Table 8-11 Total Planning Distance		
Discharge From	Distance (miles)	Evaluation Details
BFSF	15.70 [1]	A spill from this location could potentially reach the confluence of House Creek and Cowhouse Creek in approximately 7.2 hours and travel 4.8 hours (approximately 4.7 miles) on Cowhouse Creek within the 12 hour Tier 1 Response Time. This point on Cowhouse Creek is still within the boundary of Fort Cavazos.
RGAF	18.52 [2]	A spill from this location could potentially reach the confluence of Reese Creek and Lampasas River in approximately 4.4 hours and travel on Lampasas River for approximately 7.6 hours within the 12 hour response time. This means it will travel approximately 9.1 miles. The discharge leaves the boundary of Fort Cavazos approximately 5.6 miles downstream from the tank.
YDAHP RRF	18.66 [3]	A spill from this location could potentially reach the confluence of South Nolan Creek and Nolan Creek in approximately 11.0 hours and travel on Nolan Creek for approximately 1.0 hour. The discharge leaves the boundary of Fort Cavazos as soon as it enters the concrete-lined channel.
Notes: [1] Response equipment will reach the leading the spill when it is 8.6 miles from Belton Lake. [2] Response equipment will reach the spill when it is 9.5 miles from Stillhouse Hollow Lake. [3] Response equipment will reach the spill when it is on Nolan Creek.		